A Project Report

On

AGENTIC AI FOR SMART PROCUREMENT IN MANUFACTURING INDUSTRIES

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APPROVAL SHEET

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Declaration

We declare that this written submission for B.E. Declaration entitled "**Agentic AI for Smart Procurement In Manufacturing Industries**" represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any ideas/data/fact/source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have thus not been properly cited or from whom permission has not been taken when needed.

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Abstract

Manufacturing procurement involves document processing, inventory assessment, vendor selection, purchase-order generation, approvals, and risk analysis. Traditional manual and rule-based approaches can make these activities time-consuming and fragmented. This project presents an Agentic AI system for smart procurement in manufacturing industries using a modular multi-agent architecture. Specialized agents support document intelligence and Optical Character Recognition (OCR), inventory analysis, vendor evaluation, procurement processing, approval coordination, risk assessment, notifications, reporting, and vendor interaction. The system combines Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) with technologies including Next.js/React, FastAPI, PostgreSQL, Redis, Apache Kafka, Docker, and MinIO. A vendor chatbot supports queries and negotiation-related interactions, while human approval is retained for critical procurement decisions. The implementation includes purchase-order generation, vendor scoring, role-based approvals, email notifications, digital signatures, and audit tracking. Functional and integration test cases were used to validate key workflows. The project demonstrates how coordinated AI agents can reduce repetitive effort, improve traceability, and support more informed procurement decisions while retaining human oversight.

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Chapter 1

Introduction

1.1 Background

Manufacturing procurement has evolved from conventional manual and rule-based processes toward more automated and data-driven approaches. The increasing complexity of supply chains requires efficient management of inventory, suppliers, purchase orders, and procurement-related information. Advances in Artificial Intelligence, multi-agent systems, Large Language Models, and Retrieval-Augmented Generation provide opportunities to automate and improve these activities. Multi-agent architectures allow procurement tasks to be divided among specialized agents that can coordinate and exchange information during the workflow. The integration of AI-powered conversational interfaces further enables direct interaction with vendors for queries and negotiation during the procurement process. In the proposed system, these technologies are applied to document processing, inventory evaluation, vendor intelligence, procurement execution, and risk analysis. The system therefore provides a foundation for integrating intelligent automation with human decision-making in manufacturing procurement.

1.2 Motivation

The motivation for this project arises from the need to reduce manual effort and delays in manufacturing procurement processes. Traditional procurement approaches often require separate handling of documents, inventory information, vendor evaluation, approvals, risk assessment, and vendor communication. The availability of AI and multi-agent technologies provides an opportunity to automate these activities while improving coordination between different stages of the procurement workflow. The project is motivated by the need for faster vendor evaluation, efficient inventory management, automated procurement processing, and streamlined vendor interaction through an AI-powered chatbot. It also aims to retain human involvement in important approval and negotiation decisions while allowing AI agents to handle repetitive analysis and communication tasks. Therefore, the proposed system seeks to provide an integrated, intelligent, and scalable approach to procurement management.

1.3 Aim and Objective

Aim:

The aim of this project is to develop an intelligent, AI-driven procurement system for manufacturing industries using a modular multi-agent architecture. The proposed system, ProcureFlow, is designed to coordinate important procurement activities—including document processing, inventory assessment, vendor evaluation, purchase-order generation, risk analysis, approval coordination, and vendor interaction—within an integrated workflow. By combining Optical Character Recognition (OCR), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and workflow automation, the system seeks to reduce repetitive manual effort, improve traceability, and support informed procurement decisions while retaining human oversight for critical approvals.

Objectives:
The main objectives of the project are as follows:

1. To process procurement-related documents and extract relevant information using document intelligence and Optical Character Recognition (OCR).
2. To assess inventory availability and identify stock shortages so that procurement requirements can be determined more effectively.
3. To evaluate and score vendors using available vendor and procurement information, supporting the selection of suitable suppliers.
4. To coordinate procurement activities through specialized AI agents and generate purchase orders based on the identified requirements.
5. To assess procurement-related risks and support a controlled, tier-based approval workflow, keeping authorized human users responsible for critical decisions.
6. To provide an AI-powered vendor chatbot that uses Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to support vendor queries and negotiation-related interactions.
7. To support procurement workflow communication through notifications and maintain traceable records of important actions and decisions.
8. To integrate the system components through a suitable web-based architecture using technologies such as Next.js/React, FastAPI, PostgreSQL, Redis, Apache Kafka, Docker, and MinIO.
9. To evaluate key procurement workflows through functional and integration testing and assess whether the system performs its intended operations reliably.

1.4 Report Outline

This report is organized into six chapters, followed by references and supporting appendices.

- **Chapter 1 – Introduction:** Introduces the project, its background and motivation, the procurement problem being addressed, the aim and objectives, scope, assumptions, and limitations.
- **Chapter 2 – Literature Survey:** Reviews relevant research and existing approaches in multi-agent systems, supply-chain and procurement management, Optical Character Recognition, and AI-assisted decision-making. It also discusses the limitations of existing approaches, the research gap, and the system requirements.
- **Chapter 3 – Proposed System:** Describes the problem statement, scope, and proposed ProcureFlow solution. It explains how specialized agents coordinate document processing, inventory analysis, vendor evaluation, procurement processing, risk assessment, approvals, notifications, reporting, and vendor interaction.

- **Chapter 4 – Design of the System:** Presents the system architecture and design artifacts, including the procurement workflow, entity–relationship model, execution flow, and system requirements. It also describes the main components and the technologies used to connect them.
- **Chapter 5 – Results and Discussion:** Presents the implementation results and the software test cases used to evaluate the system. It discusses the observed behavior of key workflows and summarizes the overall testing results.
- **Chapter 6 – Conclusion and Future Scope:** Summarizes the project outcomes and conclusions, and identifies possible future enhancements, including payment gateway integration, improvements to the vendor chatbot, and automated quality inspection and defect tracking.

The report concludes with the references cited throughout the work and appendices containing supplementary project material, including the project timeline, Sustainable Development Goal mapping, and project outcomes.

Chapter 2

Study of the System

This chapter presents the study of the system, including the techniques used in intelligent procurement, the literature survey, existing procurement systems, research gaps, requirements analysis, and the Software Requirements Specification (SRS) for the proposed AI-driven procurement management system for manufacturing industries.

2.1 About the Technique

Research on intelligent supply-chain and procurement systems has increasingly moved from centralized approaches toward distributed and autonomous decision-making. Fu et al. [1] proposed a multi-agent framework for collaborative inventory management that integrates autonomous agents for demand forecasting, inventory planning, and replenishment. Ta et al. [2] further explored coordination among distributed manufacturers, suppliers, and distributors, emphasizing decentralized decision-making under changing conditions. Tian and Tianfield [3] reviewed multi-agent supply-chain management and identified modelling, simulation, negotiation, and coordination as major research areas. Similarly, Gao and Qi [4] developed a multi-agent logistics information system using coordination, task, and interface agents, while Zhang and Tao [5] introduced dynamic reconfiguration to allow supply-chain systems to adapt to changing customer requirements and supplier conditions. Li [6] applied multi-agent technology to an automobile-parts supply chain, integrating information sharing, procurement, demand forecasting, production planning, and quality tracking. Zhao et al. [7] combined multi-agent systems with ant-colony optimization for distributed supply-chain decision-making, while Ghadimi and Heavey [8] proposed MASOS (Multi-Agent System Simulation Framework for Sustainable Supplier Evaluation and Order Allocation) for supplier evaluation and order allocation using a multi-layer agent architecture and fuzzy inference. Hachaïchi et al. [9] investigated reinforcement learning and Proximal Policy Optimization for inventory replenishment and automated order-quantity decisions. Together, these studies establish the importance of autonomous agents, coordination, optimization, supplier evaluation, and adaptive decision-making in supply-chain and procurement environments.

The literature also highlights the importance of document intelligence and artificial intelligence in automating procurement-related activities. Memon et al. [10] systematically reviewed handwritten OCR research and identified feature extraction, classification, and deep-learning techniques for converting document images into machine-readable information. Smith [11] presented the architecture of the Tesseract OCR engine, including text-line detection, feature extraction, classification, and adaptive recognition, providing a technical foundation for extracting information from scanned procurement documents. Danach et al. [12] examined the broader application of AI in supply-chain management, including predictive analytics, demand forecasting, inventory optimization, logistics, and automation. Xia [13] investigated the integration of LLM agents with digital twins and proposed structured representations and multi-agent task decomposition for autonomous industrial processes. ChatSync by Li et al. [14] demonstrated how LLM reasoning can be combined with structured spatial-temporal knowledge to support logistics synchronization and traceability. Jannelli et al. [15] extended this direction through LLM-based multi-agent communication, negotiation, and consensus-seeking in supply-chain decision-making. Wang et al. [16] demonstrated the use of Retrieval-Augmented Generation (RAG) for incorporating domain-specific supply-chain knowledge into LLM reasoning. Jiang and Hu [17] similarly combined specialized agents, RAG, LLM-based reasoning, and domain knowledge in an agentic architecture, although their application was predictive maintenance rather than procurement. Maheshwari et al. [18] directly addressed

purchase-order automation by combining forecasting models with automated PO generation for semiconductor supply chains. Henkel et al. [19], through a systematic review of foundation-model-based industrial agents, further identified challenges such as hallucination, generalization, output instability, data scarcity, and inference latency, highlighting the importance of validation and human oversight in agentic systems.

Recent studies are increasingly aligned with intelligent supplier selection, procurement reasoning, and end-to-end agentic workflows. Gao et al. [20] proposed SCSimulator, an LLM-driven multi-agent framework for supply-chain partner selection that combines multi-criteria decision-making, game-theoretic reasoning, explainable AI, and human-in-the-loop interaction. Long et al. [21] introduced HELICASE, which uses autonomous LLM agents and uncertainty-guided investigation to construct supply-chain knowledge graphs and support evidence-based reasoning, making it relevant to supplier discovery, traceability, and risk analysis. Contract2Plan [22] proposed a verified RAG-to-optimization approach in which contractual information such as minimum order quantities and lead times is retrieved, converted into constraints, and verified before generating procurement decisions. Bandara et al. [23] presented Flowr, an agentic AI framework covering demand forecasting, procurement, supplier coordination, and inventory replenishment through specialized agents, domain-specific LLMs, orchestration, and human-in-the-loop validation. These studies collectively indicate a shift toward procurement systems that combine multi-agent collaboration, RAG-based knowledge grounding, document intelligence, supplier evaluation, risk analysis, verification, and human oversight. However, many existing research works address only individual stages or specific use cases rather than integrating these capabilities into a unified procurement workflow.

2.2 Various Available Technique

2.2.1 Multi-Agent Systems

Research on multi-agent systems has addressed collaborative inventory management, supply-chain coordination, distributed decision-making, supplier evaluation, and order allocation. These systems use autonomous agents to perform specialized tasks and coordinate their activities across different supply-chain participants. Existing studies also explore dynamic reconfiguration, ant-colony optimization, and reinforcement learning for improving supply-chain decisions.

2.2.2 Optical Character Recognition

Optical Character Recognition (OCR) is used to convert document-based information into machine-readable text. Memon et al. [10] reviewed OCR techniques involving feature extraction, classification, and deep learning. Smith [11] described the Tesseract OCR engine, including text-line detection, feature extraction, classification, and adaptive recognition. These techniques provide a foundation for extracting information from scanned procurement documents.

2.2.3 Artificial Intelligence and Large Language Models

Artificial intelligence techniques are applied in predictive analytics, demand forecasting, inventory optimization, logistics, and supply-chain automation. Large Language Model (LLM) agents extend these capabilities through reasoning, planning, and multi-agent task

decomposition. Research has also explored their integration with digital twins, structured knowledge, logistics synchronization, communication, negotiation, and consensus-seeking.

2.2.4 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) incorporates domain-specific information into LLM-based reasoning. Existing research explores RAG for supply-chain knowledge, specialized agent architectures, and contractual information retrieval. Contract2Plan [22], for example, retrieves contractual information such as minimum order quantities and lead times, converts it into constraints, and verifies the constraints before generating procurement decisions.

2.2.5 Supplier Evaluation and Procurement Automation

Existing approaches address supplier evaluation, partner selection, purchase-order generation, inventory planning, and order allocation. Some systems combine forecasting models with automated purchase-order generation, while others use multi-agent simulation, multi-criteria decision-making, and explainable AI for partner selection. These techniques demonstrate the use of AI for improving procurement-related decisions.

2.2.6 Limitations of Available Techniques

The literature identifies challenges involving hallucination, generalization, output instability, data scarcity, inference latency, and the need for human oversight. OCR techniques focus on document recognition rather than their integration into procurement workflows. Several multi-agent and LLM-based approaches address specific activities, such as supplier selection, inventory planning, knowledge discovery, or retail operations, instead of the complete procurement lifecycle.

2.3 Related Works

2.3.1 Literature Survey and Analysis

The following table reviews various studies related to AI-driven procurement systems designed to automate procurement processes and improve decision-making in manufacturing industries.

2.3.1.1 Review of Related Work

Ref.	Paper Title	Key Features	Limitations
[1]	**Multi-Agent Enabled Modeling and Simulation Towards Collaborative Inventory Management in Supply Chains**	Employs multiple autonomous agents for collaborative inventory management, demand forecasting, planning, and replenishment.	Primarily simulation-oriented and does not provide a complete implementation of an integrated procurement system.

Ref.	Paper Title	Key Features	Limitations
[2]	**Study on Coordination in Multi-Agent-Based Agile Supply Chain Management**	Uses autonomous agents to coordinate manufacturers, suppliers, and distributors under changing supply-chain conditions.	Primarily addresses inter-agent coordination; practical implementation and deployment aspects are limited.
[3]	**Literature Review upon Multi-Agent Supply Chain Management**	Reviews the application of multi-agent systems to supply-chain modelling, negotiation, coordination, and simulation.	Highlights challenges related to simulation complexity and coordination among heterogeneous agents.
[4]	**Study on Cooperative Logistics Information System According to Supply Chain Management**	Introduces coordination, task, and interface agents to support information sharing and distributed logistics operations.	Requires further improvement in coordination mechanisms and validation through real-world deployment.
[5]	**Multi-Agent Based Supply Chain Management with Dynamic Reconfiguration Capability**	Provides dynamic reconfiguration capabilities that allow agents to adapt to changes in demand, suppliers, and operating conditions.	Evaluation is primarily simulation-based and does not address a complete intelligent procurement workflow.
[6]	**Study on E-Supply Chain Management in Automobile Part Industry Based on Multi-Agent System**	Integrates information sharing, procurement, demand forecasting, production planning, and supply-chain coordination using multi-agent technology.	Designed specifically for the automobile-parts domain, limiting direct applicability to general procurement environments.
[7]	**Research on Supply Chain Management Using Multi-Agent System Based on Ant Colony Algorithm**	Combines multi-agent systems with ant-colony optimization for distributed supply-chain decision-making and path selection.	Focuses mainly on mathematical optimization and does not provide comprehensive procurement automation.
[8]	**MASOS: A Multi-Agent System Simulation Framework for Sustainable Supplier Evaluation and Order Allocation**	Uses multiple agents and fuzzy inference to evaluate suppliers and allocate orders across multiple suppliers.	Framework-level research requiring further implementation, validation, and evaluation in practical procurement environments.
[9]	**A Policy Gradient Based Reinforcement Learning Method for Supply Chain Management**	Applies reinforcement learning and policy-gradient methods to optimize inventory replenishment and future order quantities.	Primarily addresses inventory ordering and replenishment rather than the complete procurement lifecycle.

Ref.	Paper Title	Key Features	Limitations
[10]	**Handwritten Optical Character Recognition (OCR): A Comprehensive Systematic Literature Review**	Reviews OCR techniques for converting handwritten and document-based information into machine-readable text using feature extraction, classification, and deep learning.	Focuses on OCR techniques rather than their integration into procurement or supply-chain workflows.
[11]	**An Overview of the Tesseract OCR Engine**	Describes the Tesseract OCR pipeline, including text-line detection, feature extraction, classification, and adaptive recognition.	OCR accuracy may vary depending on document quality and can be lower than some advanced commercial OCR solutions.
[12]	**Revolutionizing Supply Chain Management With AI: A Path to Efficiency and Sustainability**	Explores the use of AI for predictive analytics, demand forecasting, inventory optimization, logistics, and supply-chain automation.	Provides a broad overview of AI in supply chains rather than a specific multi-agent procurement architecture.
[13]	**Integrating Large Language Model Agents with Digital Twins for Industrial Autonomous Systems**	Combines LLM agents, digital twins, task decomposition, planning, and autonomous decision-making for industrial applications.	Reliability, task complexity, output consistency, and the requirement for human validation remain significant challenges.
[14]	**ChatSync: Large-Language-Model-Enabled Spatial–Temporal Knowledge Reasoning for Production Logistics Synchronization**	Combines LLM-based reasoning with structured spatial-temporal knowledge to support production logistics coordination, resource allocation, and traceability.	Primarily focuses on production logistics synchronization rather than procurement management.
[15]	**Agentic LLMs in the Supply Chain: Towards Autonomous Multi-Agent Consensus-Seeking**	Uses LLM-based agents for communication, negotiation, consensus-seeking, and supply-chain decision-making.	Potential hallucinations, tool reliability, prompt sensitivity, and the need for human oversight remain concerns.
[16]	**LLMs for Supply Chain Management**	Uses Retrieval-Augmented Generation (RAG) to incorporate external domain-specific supply-chain knowledge into LLM-based reasoning.	Knowledge-base scalability and broader evaluation across complex supply-chain scenarios require further investigation.

Ref.	Paper Title	Key Features	Limitations
[17]	**Artificial Intelligence Agent-Enabled Predictive Maintenance: Conceptual Proposal and Basic Framework**	Combines AI agents, RAG, LLM-based reasoning, and digital twins to support automated maintenance-related decisions.	The proposed framework targets predictive maintenance rather than procurement and supplier management.
[18]	**Forecast Accuracy Improvement and Purchase Order Automation Using a Multi-Model Approach in Semiconductor Supply Chain: A Case Study**	Combines forecasting models with automated purchase-order generation to improve material planning and procurement efficiency.	Does not comprehensively address supplier risk, logistics risk, supplier-specific policies, or broader procurement activities.
[19]	**Foundation-Model-Based Agents in Industrial Automation: Purposes, Capabilities, and Open Challenges**	Reviews foundation-model-based agents for industrial automation and identifies their capabilities, applications, and maturity.	Identifies continuing challenges involving hallucination, reliability, data availability, generalization, and inference time.
[20]	**SCSimulator: An Exploratory Visual Analytics Framework for Partner Selection in Supply Chains through LLM-Driven Multi-Agent Simulation**	Uses LLM-driven multi-agent simulation, multi-criteria decision-making, explainable AI, and human interaction for supply-chain partner selection.	Primarily focuses on partner selection and simulation rather than an end-to-end procurement workflow.
[21]	**Helicase: Uncertainty-Guided Supply Chain Knowledge Graph Construction with Autonomous Multi-Agent LLMs**	Uses autonomous multi-agent LLMs to collect, verify, organize, and construct supply-chain knowledge through uncertainty-guided investigation.	Primarily addresses supply-chain knowledge discovery and construction rather than procurement execution.
[22]	**Verified Contract-Grounded Inventory Planning**	Uses RAG and verification to retrieve contractual information, extract procurement constraints, and support evidence-grounded inventory decisions.	Focuses on contract-grounded inventory planning rather than the complete procurement lifecycle.
[23]	**Flowr: Scaling Up Retail Supply Chain Operations Through Agentic AI in Large Scale Supermarket Chains**	Uses specialized AI agents for demand forecasting, procurement, supplier coordination, and inventory management with human oversight.	Primarily targets large-scale retail environments and retains human involvement for important decisions.

Table 1: Summary of Reviewed Literature

2.3.2 Existing System

In addition to academic research, several commercial procurement platforms demonstrate the practical adoption of digital source-to-pay and supplier-management capabilities. Coupa provides a Business Spend Management platform covering procurement, sourcing, supplier management, purchasing, invoicing, payments, spend analysis, and supply-chain planning, with supplier collaboration supported through its Supplier Portal [24]. Ivalua provides an integrated procurement platform connecting workflows, data, AI agents, supplier management, sourcing, purchasing, and governance; its current platform also supports agentic workflows and controlled AI autonomy across source-to-pay processes [25]. JAGGAER offers an intelligent source-to-pay platform covering source-to-contract, procure-to-pay, supplier intelligence, sourcing, contracts, purchase orders, invoicing, payments, and supplier performance and risk management [26]. Oracle Fusion Cloud Procurement provides integrated procurement capabilities including sourcing, purchasing, supplier qualification and management, contracts, supplier portals, requisitions, purchase orders, and procure-to-pay processes; Oracle also currently incorporates AI and agentic applications into its procurement offering [27]. These systems demonstrate that modern enterprise procurement platforms already provide extensive workflow automation, supplier collaboration, sourcing, purchasing, and spend-management capabilities. However, the research literature shows continued opportunities for modular multi-agent architectures that explicitly combine document/OCR processing, RAG-grounded reasoning, supplier intelligence, risk analysis, verification, and human-in-the-loop decision-making within a single procurement workflow. The proposed system therefore builds upon both the academic multi-agent literature and the capabilities demonstrated by existing enterprise procurement platforms while focusing on an integrated, AI-driven procurement architecture.

2.3.2.1 Review of Existing System

This table presents existing procurement systems and their key features and limitations in comparison with the proposed AI-driven procurement system.

Existing System	Main Features	Limitations / Gap Relative to Proposed System
Coupa	Provides supplier portals, catalog management, purchase orders, invoicing, and integrated procurement workflows.	Primarily emphasizes procurement transactions; advanced manufacturing-specific processes may require integration with additional systems.
Ivalua	Provides supplier management, strategic sourcing, bill of materials (BOM) management, risk and quality management, and procurement capabilities.	Provides extensive functionality, but configuring and managing multiple modules can be resource-intensive for smaller organizations.

Existing System	Main Features	Limitations / Gap Relative to Proposed System
JAGGAER	Supports sourcing, BOM-based procurement, supplier risk management, inventory management, and supplier quality management.	Provides strong manufacturing support, but integration with existing ERP and plant-level systems may be required.
Oracle Procurement	Provides supplier management, sourcing, purchasing, contract management, and procurement approval workflows.	Strongly integrated with the broader Oracle ecosystem, which may make integration with non-Oracle systems more challenging.

Table 2: Comparison of Existing Procurement Systems

2.3.3 Research Gap and Limitations

The literature shows a clear gap: existing research addresses individual parts of the procurement and supply-chain process, such as inventory management, forecasting, supplier evaluation, OCR-based document processing, RAG, LLM reasoning, risk/knowledge analysis, and purchase-order automation, but these capabilities are generally not integrated into one end-to-end procurement system. Traditional multi-agent approaches provide coordination but lack modern LLM-based reasoning, while recent LLM-agent studies demonstrate intelligent decision-making but often focus on specific tasks such as supplier selection, knowledge discovery, inventory planning, or retail operations.

Our proposed system overcomes this gap by integrating specialized AI agents into a single event-driven procurement workflow, where documents are processed using OCR, inventory and requirements are analysed, suppliers are evaluated using relevant data and RAG-based evidence, risks are assessed, decisions are coordinated across agents, and purchase orders are generated and routed through an approval process. This provides a more automated, modular, evidence-based, and human-supervised procurement solution than the individual approaches presented in the existing literature.

2.3.4 Requirements Analysis

The requirement analysis for the proposed AI-driven procurement management system focuses on identifying the functional and non-functional requirements necessary to develop an efficient, scalable, and intelligent procurement solution for manufacturing industries. The system aims to automate major procurement activities such as document processing, inventory analysis, vendor evaluation, purchase-order generation, and approval support while maintaining human involvement in critical decisions.

2.3.4.1 Functional Requirements

The system should be able to accept and process procurement documents and extract relevant information using OCR and document-processing techniques. It must analyze inventory data to identify available quantities and shortages and evaluate suitable vendors based on factors such as price, rating, and delivery performance. The system should generate procurement requirements and purchase orders for selected vendors and support the approval workflow. It should provide AI/RAG-based analysis for vendor and risk assessment and include a vendor chatbot for handling queries, supporting communication,

and facilitating price negotiation. The chatbot may also use ML-based budget or price prediction to provide a reference range during negotiations.

2.3.4.2 Non-Functional Requirements

The system should provide efficient response times and reliable communication between its distributed components. It should be scalable to accommodate increasing procurement data, users, and chatbot interactions while maintaining consistent performance. Data security and privacy should be maintained for procurement, inventory, vendor, and transaction information. The system should provide reliable and consistent AI-generated responses, with human validation for critical procurement and negotiation decisions. The interface should be user-friendly for both internal procurement users and vendors, while the system should remain maintainable and deployable across environments using containerization technologies such as Docker.

2.3.5 Software Requirements Specification (SRS)

2.3.5.1 Introduction

Purpose This document specifies the requirements of ProcureFlow, an AI-driven procurement management system for manufacturing industries.

Scope The system automates document processing, inventory analysis, vendor evaluation, purchase order generation, risk assessment, and approval workflows. It also supports vendor queries and price negotiation through an AI-powered chatbot.

Definitions and Abbreviations

- **AI:** Artificial Intelligence
- **LLM:** Large Language Model
- **RAG:** Retrieval-Augmented Generation
- **OCR:** Optical Character Recognition
- **PO:** Purchase Order
- **SRS:** Software Requirements Specification
- **SMTP:** Simple Mail Transfer Protocol

2.3.5.2 Overall Description

Product Perspective ProcureFlow is a web-based platform using AI agents, FastAPI, PostgreSQL, Redis, and Apache Kafka to coordinate procurement activities.

Product Functions

- Procurement document processing using OCR and AI.
- Inventory availability and shortage analysis.
- Vendor evaluation and risk assessment.
- Purchase order generation and approval.
- SMTP-based email notifications.
- AI-powered vendor chatbot.
- Procurement tracking and reporting.

User Classes

- **Procurement User:** Initiates and monitors procurement.
- **Approver:** Reviews and approves procurement requests.
- **Vendor:** Interacts with the chatbot for queries and negotiation.
- **Administrator:** Manages system configuration and access.

Operating Environment

- **Operating System:** Windows/Linux
- **Frontend:** Next.js/React
- **Backend:** Python/FastAPI
- **Database:** PostgreSQL
- **Messaging and Caching:** Apache Kafka and Redis
- **Deployment:** Docker

Constraints

- Depends on the quality of inventory and vendor data.
- AI-generated results may require validation.
- External AI and email services must be available when required.
- Critical procurement decisions require human approval.

2.3.5.3 Functional Requirements

The system shall:

1. Accept and process procurement documents.
2. Extract relevant information using OCR and AI.
3. Identify inventory shortages.
4. Evaluate vendors using price, rating, and delivery performance.
5. Generate purchase orders for selected vendors.
6. Support procurement risk analysis and approval workflows.
7. Send approval notifications through SMTP.
8. Provide chatbot-based vendor interaction.
9. Store procurement and transaction information.

2.3.5.4 Non-Functional Requirements

- **Performance:** Process procurement requests efficiently.
- **Security:** Protect procurement and vendor data from unauthorized access.
- **Scalability:** Support increasing data and transaction volumes.
- **Reliability:** Handle service delays and failures appropriately.
- **Usability:** Provide a simple, user-friendly interface.
- **Maintainability:** Support modular updates and integration.
- **Auditability:** Maintain records of important procurement decisions.

2.3.5.5 Hardware Requirements

- **Processor:** Dual-core or equivalent.
- **RAM:** Minimum 8 GB recommended for development.
- **Storage:** Sufficient space for application files, datasets, and databases.
- **Network:** Connectivity for required external services.

2.3.5.6 Software Requirements

- **Frontend:** Next.js / React
- **Backend:** Python / FastAPI
- **Database:** PostgreSQL
- **Caching:** Redis
- **Messaging:** Apache Kafka
- **AI Technologies:** LLM, RAG, OCR
- **Email:** SMTP service
- **Deployment:** Docker
- **Tools:** VS Code, Git, GitHub

2.3.5.7 System Design Overview

- **Frontend:** Provides the procurement user interface.
- **Backend:** Coordinates requests through FastAPI services.
- **AI Agents:** Handle document processing, inventory analysis, vendor evaluation, procurement, and risk analysis.
- **Database:** PostgreSQL stores persistent procurement data.
- **Kafka and Redis:** Support asynchronous communication and caching.
- **Vendor Chatbot:** Supports vendor queries and price negotiation.
- **SMTP:** Sends approval notifications.

2.3.5.8 Use Case Summary

1. User submits a procurement request or document.
2. The system extracts relevant information.
3. Inventory availability and shortages are checked.
4. Suitable vendors are evaluated.
5. A purchase order is generated.
6. An approval request is sent through email.
7. The authorized user reviews the request.
8. The system updates the procurement workflow.

2.3.5.9 Assumptions and Dependencies

- Procurement, inventory, and vendor data are available.
- Required backend services and databases are operational.
- External AI and email services are accessible when needed.
- Authorized users review critical procurement decisions.
- Recommendation accuracy depends on data quality and AI performance.

Chapter 3

Proposed System

3.1 Problem Statement

Manufacturing organizations often face significant challenges in managing procurement due to multiple manual, time-consuming, and disconnected activities. They struggle to efficiently process procurement documents, evaluate inventory availability, identify suitable vendors, generate purchase orders, assess procurement risks, and manage communication with vendors. Traditional procurement systems are largely rule-based and provide limited intelligent decision-making and automated vendor interaction, making it difficult to respond efficiently to changing requirements and supplier conditions. As a result, procurement operations may experience delays, increased manual effort, inefficient vendor selection, and fragmented communication. Therefore, there is a need for an intelligent, integrated, and data-driven procurement system that can automate key procurement activities through specialized AI agents, support vendor queries and negotiation through a chatbot, and retain human oversight for critical decisions.

3.2 Scope

The proposed system focuses on automating key manufacturing procurement activities, including document processing, inventory availability analysis, vendor evaluation, purchase-order generation, risk assessment, approval support, and vendor interaction through an AI-powered chatbot. The chatbot is intended to handle vendor queries and support price negotiation, with ML-based budget or price prediction providing additional decision support. The system assumes that procurement documents, inventory records, and vendor information are available in a usable digital format and that the required AI/LLM and external services are accessible when needed. The system is designed to support, rather than completely replace, human decision-making, with human users responsible for reviewing and approving critical procurement and negotiation decisions.

3.3 Proposed System

The proposed system uses an AI-powered multi-agent architecture to coordinate document processing, inventory analysis, vendor evaluation, procurement processing, approval management, risk assessment, notifications, reporting, and vendor interaction. It integrates OCR, LLM-based reasoning, Retrieval-Augmented Generation (RAG), and workflow automation while retaining human approval for critical procurement decisions. The implementation uses technologies including Next.js/React, FastAPI, PostgreSQL, Redis, Apache Kafka, Docker, and MinIO.

Chapter 4

Design of the System

4.1 Requirement Engineering

4.1.1 Requirement Analysis

The system requires a combination of computing resources, backend and frontend runtimes, distributed middleware, databases, and AI-based document processing services. The minimum configuration is sufficient for development and demonstration, while the recommended configuration is intended for a production-like deployment.

4.1.1.1 UML diagrams/ DFDs based on the project

The ER diagram represents the database structure of the Agentic AI Manufacturing Procurement System, illustrating the key entities, their attributes, and relationships. It covers procurement request management, document processing, purchase-order tracking, vendor management, inventory records, and vendor risk analysis.

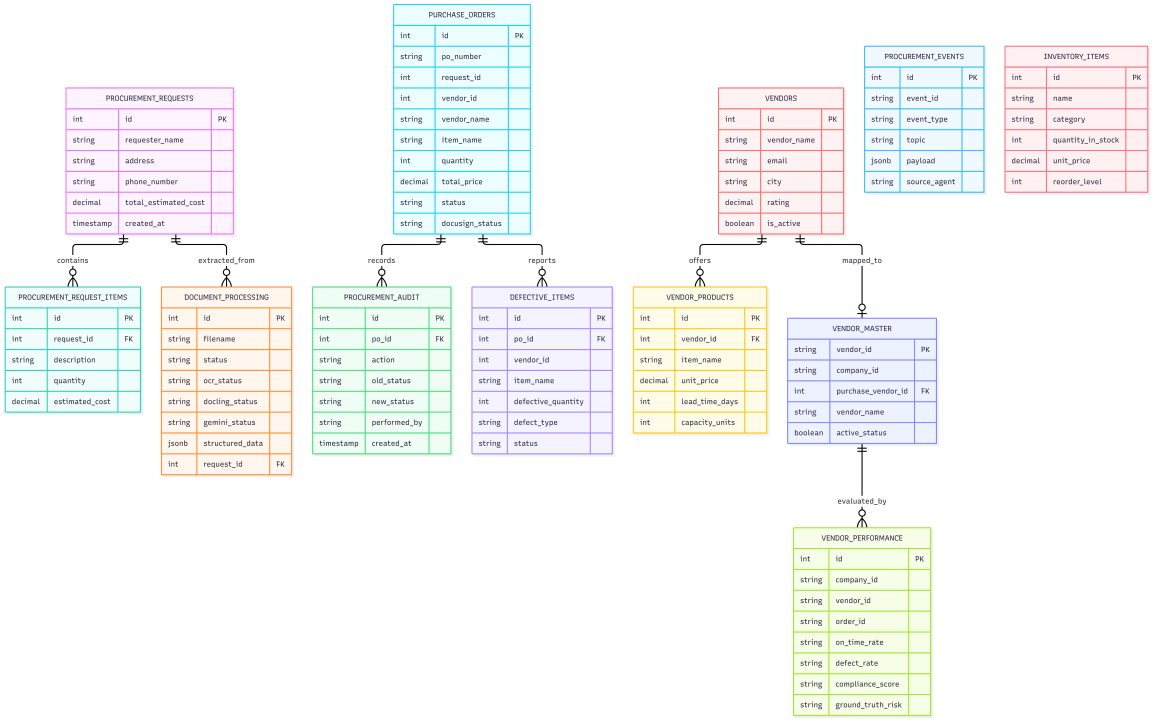


Figure 4.1: Entity–Relationship Diagram of the Proposed System

The `PROCUREMENT_REQUESTS` and `PROCUREMENT_REQUEST_ITEMS` entities store procurement requirements and their associated items. The `DOCUMENT_PROCESSING` entity tracks uploaded documents and their OCR processing status. The `PURCHASE_ORDERS` entity maintains purchase-order details and approval status, while `PROCUREMENT_AUDIT`, `PROCUREMENT_EVENTS`, and `DEFECTIVE_ITEMS` support audit tracking, event monitoring, and defect management.

The `VENDORS` and `VENDOR_PRODUCTS` entities maintain supplier and product information, including pricing and lead times. `INVENTORY_ITEMS` stores stock-related information for the supported industries. The `VENDOR_MASTER` and `VENDOR_PERFORMANCE` entities support vendor identification, performance evaluation, and risk analysis.

Overall, the database design organises procurement data into related entities while maintaining separation between OCR records, procurement transactions, and industry-

specific data sources. This structure supports data management, traceability, and informed procurement decisions across the multi-agent system.

4.1.1.2 Cost Analysis

The computational complexity of the system depends mainly on the number of procurement line items and the number of qualified vendors considered during vendor evaluation.

Computational Complexity For a requisition containing M line items, inventory verification is performed for each item and therefore has a time complexity of:

$$O(M)$$

Redis caching is used for frequently accessed inventory and vendor information, providing approximately $O(1)$ lookup time for cached data.

The vendor evaluation process considers multiple qualified vendors for each line item. If V represents the number of qualified vendors, the overall vendor evaluation complexity is:

$$O(M \times V)$$

In the standard industrial catalog configuration, the number of qualified vendors is limited to approximately 20 per line item. Therefore, the vendor ranking operation can be completed in under 10 milliseconds under the evaluated configuration.

Database operations use B-Tree indexed foreign keys, providing approximately:

$$O(\log N)$$

lookup performance for indexed database operations, where N represents the number of relevant database records.

Operational Cost The system uses PostgreSQL, MinIO, Redis, and Apache Kafka in Docker containers. Self-hosting these services on local or on-premises infrastructure keeps recurring infrastructure costs low.

LLM inference is the primary variable operational cost because document processing requires AI-based structured extraction. The project evaluation reports an average inference cost of less than Rs. 0.05 per invoice using the selected Groq or Gemini-based processing configuration. Compared with manual data-entry outsourcing, the evaluated configuration provides a reported cost reduction of more than 98%.

4.1.1.3 Hardware and software requirements

The backend services are implemented using Python and FastAPI, providing lightweight APIs for the individual agents. The frontend is developed using Next.js 14 and React to provide an interface for document upload, inventory monitoring, vendor recommendations, purchase orders, and approval tracking. Apache Kafka provides asynchronous communication between the agents, while Redis is used for frequently accessed information such as inventory and vendor data.

PostgreSQL is used for structured procurement and master-data storage. MinIO provides S3-compatible object storage for uploaded invoices and generated purchase-order documents. Tesseract OCR is used for text extraction from procurement documents, while Groq Llama-3.3-70b and Gemini 1.5 Flash provide LLM-based structured extraction and fallback processing.

Table 4.1: System Requirements

Category	**Minimum Specification**	**Production Recommendation**
Processor	Intel Core i5 / AMD Ryzen 5, 4 cores / 8 threads	8-core Intel Xeon / AMD EPYC
System Memory	8 GB RAM	16–32 GB DDR4/DDR5
Disk Storage	20 GB free SSD storage	100 GB NVMe SSD for persistent documents
Operating System	Ubuntu 22.04 LTS / Debian 12 / Windows WSL2	Ubuntu Linux 22.04 LTS Server
Backend Runtime	Python 3.10/3.11, FastAPI, Uvicorn, SQLAlchemy	Python 3.11 with Gunicorn/Uvicorn workers
Frontend Framework	Node.js 18/20, Next.js 14, React, Tailwind CSS	Next.js 14 standalone production deployment
Middleware and Queues	Apache Kafka 7.4, Zookeeper, Redis 7	Clustered multi-broker Kafka with Schema Registry
AI / ML Services	Tesseract OCR v5, Groq Llama-3.3-70b, Gemini 1.5 Flash	Enterprise Groq API or local vLLM host

4.2 System architecture

The architecture of the proposed Agentic AI Manufacturing Procurement System is designed to automate procurement operations through a modular, multi-agent framework. As illustrated in Figure 4.2, the system is organised into three primary layers: the User Interface Layer, the Application/Agent Layer, and the Data Layer. These layers work together to support document processing, inventory assessment, vendor evaluation, purchase-order management, approval workflows, risk analysis, vendor communication, and reporting.

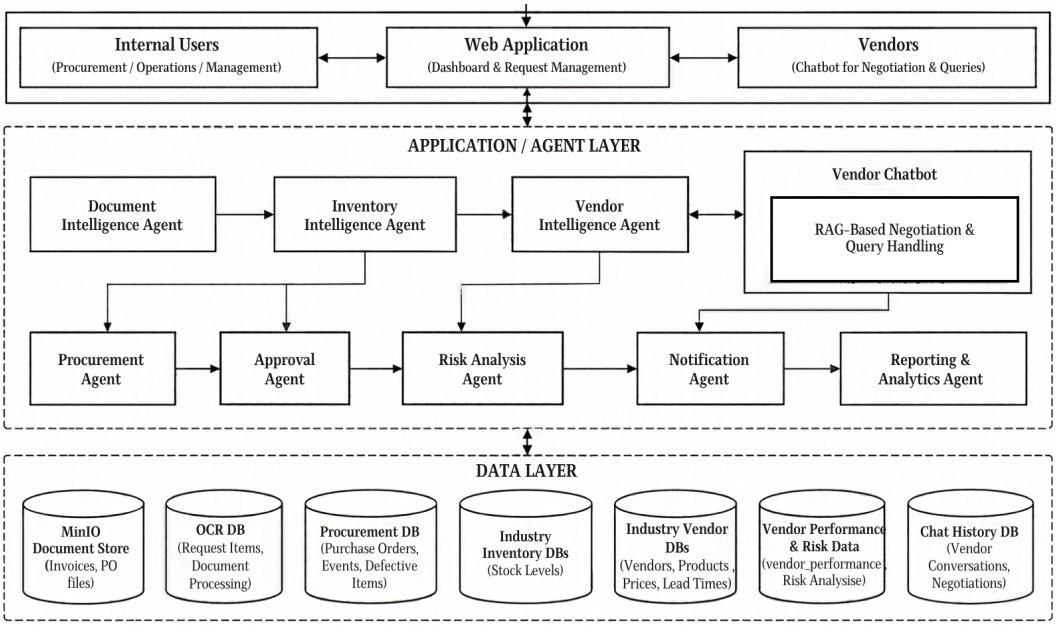


Figure 4.2: Architectural Diagram

4.2.1 UI/UX diagram

The User Interface Layer provides the interaction point between internal users, vendors, and the system. Internal users, including procurement, operations, and management personnel, access the web application to manage procurement requests and monitor workflow activities. Vendors interact with the system through the vendor chatbot to submit queries and participate in negotiation-related activities. The web application provides a unified interface for accessing procurement functionalities and relevant information.

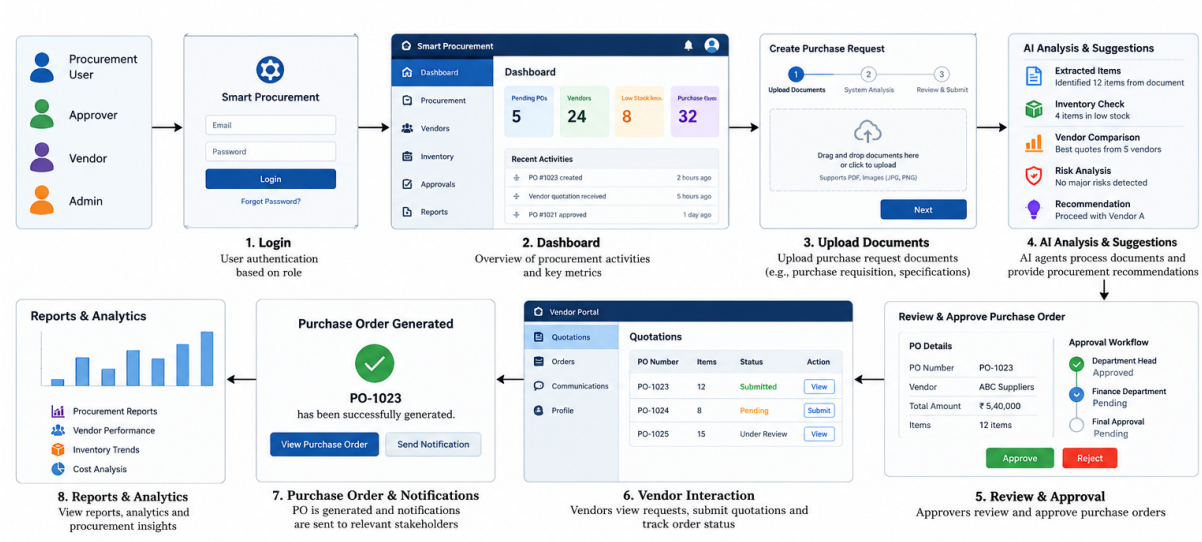


Fig. X.X: UI/UX Diagram of the Proposed System

Figure 4.3: UI/UX Diagram

4.2.2 Block Diagram

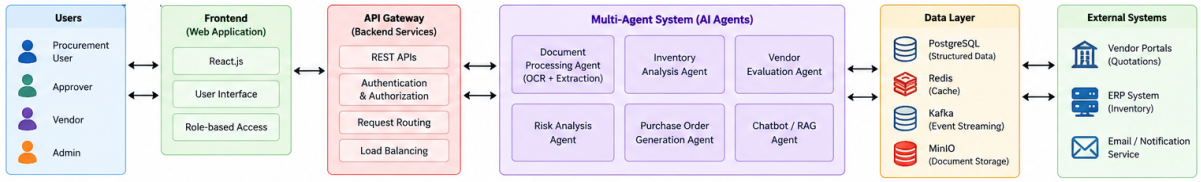


Fig. X.X: Block Diagram of the Proposed System

Figure 4.4: Block Diagram of the Proposed System

4.2.2.1 Application / Agent Layer

The Application/Agent Layer forms the core of the system and consists of specialised agents that perform individual tasks within the procurement workflow.

- **Document Intelligence Agent:** Processes uploaded procurement documents, such as invoices, purchase orders, and quotations. It extracts relevant information using OCR and document-processing techniques, validates the extracted fields, and converts unstructured content into structured procurement data.
- **Inventory Intelligence Agent:** Evaluates current inventory levels, considers reserved stock, calculates available quantities, and identifies shortages. The resulting information helps determine whether additional materials or components need to be procured.
- **Vendor Intelligence Agent:** Retrieves vendor and product information, compares prices and quality indicators, evaluates historical performance, and recommends suitable vendors based on procurement requirements.
- **Procurement Agent:** Manages procurement request processing and purchase-order generation. It prepares purchase orders containing the required item details, quantities, pricing, and selected vendor information, and supports purchase-order status tracking.
- **Approval Agent:** Coordinates the approval workflow by routing purchase orders to the appropriate approvers, checking applicable budgetary and policy requirements, collecting approval decisions, and updating approval statuses.
- **Risk Analysis Agent:** Evaluates procurement-related risks by analysing vendor performance, pricing variations, delivery reliability, and other relevant risk indicators. Its assessment supports informed vendor selection and procurement decisions.
- **Notification Agent:** Uses SMTP to send purchase-order approval emails to designated approvers according to the hierarchical approval structure, ensuring that approval requests are routed to the appropriate authority.
- **Reporting and Analytics Agent:** Consolidates procurement information to generate reports, performance indicators, and analytical insights. It supports the monitoring of procurement activities and helps identify trends and opportunities for operational improvement.

- **Vendor Chatbot:** Provides an interactive interface for vendor queries and negotiation-related activities. It incorporates Retrieval-Augmented Generation (RAG)-based negotiation and query handling to generate contextually relevant responses using retrieved information.

The agents collaborate to execute their assigned responsibilities while maintaining a modular structure that facilitates maintenance and future enhancements.

4.2.2.2 Data Layer

The Data Layer provides persistent storage for procurement documents, extracted requests, inventory information, vendor records, purchase orders, risk-related data, and other operational information. It consists of the following logical data stores:

- **MinIO Document Store:** Stores procurement documents and related files, including invoices, quotations, and purchase-order documents. It provides a central location for document storage and retrieval.
- **OCR Database:** Stores structured procurement requests, individual request items, and document-processing information, including processing statuses and extracted data. It supports the Document Intelligence Agent by maintaining the results of document extraction.
- **Procurement Database:** Maintains purchase orders, procurement audit records, procurement events, and defective-item reports. It supports purchase-order tracking, status updates, event recording, and auditability throughout the procurement lifecycle.
- **Industry Inventory Databases:** Store inventory information for the supported industries, including construction and manufacturing. These databases maintain item details, stock quantities, categories, pricing, and warehouse-related information required for inventory evaluation.
- **Industry Vendor Databases:** Maintain vendor master records and product catalogues, including supplier details, supported products, prices, lead times, minimum order quantities, and availability information. The system accesses these records through the relevant service APIs.
- **Vendor Performance and Risk Data:** Stores vendor mappings and historical performance records used for vendor evaluation and risk analysis. The `vendor_master` and `vendor_performance` tables support vendor identification and the assessment of factors such as delivery performance, defects, compliance, and other risk indicators.

The separation of these data stores reflects the system's database organisation and supports controlled access through application services and APIs. The OCR and procurement databases maintain distinct responsibilities, while the industry-specific databases provide the inventory and vendor information required for procurement decisions.

4.2.3 Flow Chart

System flowchart

The flowchart of the Agentic AI Manufacturing Procurement System illustrates the sequence of activities involved in automating procurement, from document processing to order fulfilment and reporting, through specialised agents. As shown in Figure 4.5, the workflow integrates multiple specialised agents to facilitate procurement automation, risk assessment, approval management, and reporting.

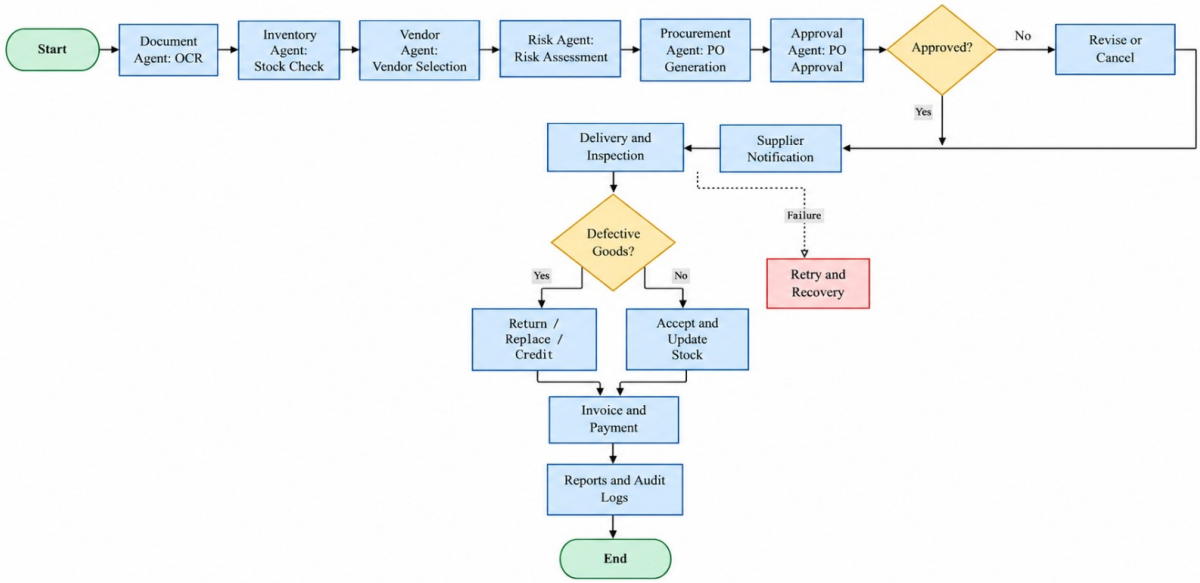


Figure 4.5: Flowchart Of Procurement system

4.2.3.1 Document Processing and Inventory Assessment

The process begins when a user uploads a procurement document through the web application. The Document Intelligence Agent uses OCR to extract and structure relevant information. The Inventory Intelligence Agent then checks stock availability and identifies shortages requiring procurement.

4.2.3.2 Vendor Selection and Risk Assessment

The Vendor Intelligence Agent evaluates suppliers based on product availability, pricing, quality, and historical performance. The Risk Analysis Agent assesses vendor and procurement risks to support informed supplier selection.

4.2.3.3 Purchase Order Generation and Approval

The Procurement Agent generates the purchase order using the procurement requirements and selected vendor details. The Approval Agent routes it through the hierarchical approval workflow. If the order is rejected, it may be revised or cancelled; if approved, the supplier is notified.

4.2.3.4 Vendor Chatbot and Notifications

The Vendor Chatbot uses Retrieval-Augmented Generation (RAG)-based negotiation and query handling to provide contextually relevant responses to vendor queries. The Notification Agent uses SMTP to send approval emails to designated approvers according to the hierarchical approval structure.

4.2.3.5 Delivery, Inspection, and Reporting

Following supplier fulfilment, the delivered goods are inspected for defects. Defective items may be returned, replaced, or resolved through credit, while accepted goods proceed to stock updates. Invoice and payment-related activities follow, and the Reporting and Analytics Agent generates reports. Procurement records and audit logs support traceability and monitoring.

4.2.4 Execution Flow / Working of System

The system execution flow is organised into three levels: the Frontend Layer, API Layer, and Backend Layer, illustrating the interaction between users, APIs, agents, and data stores.

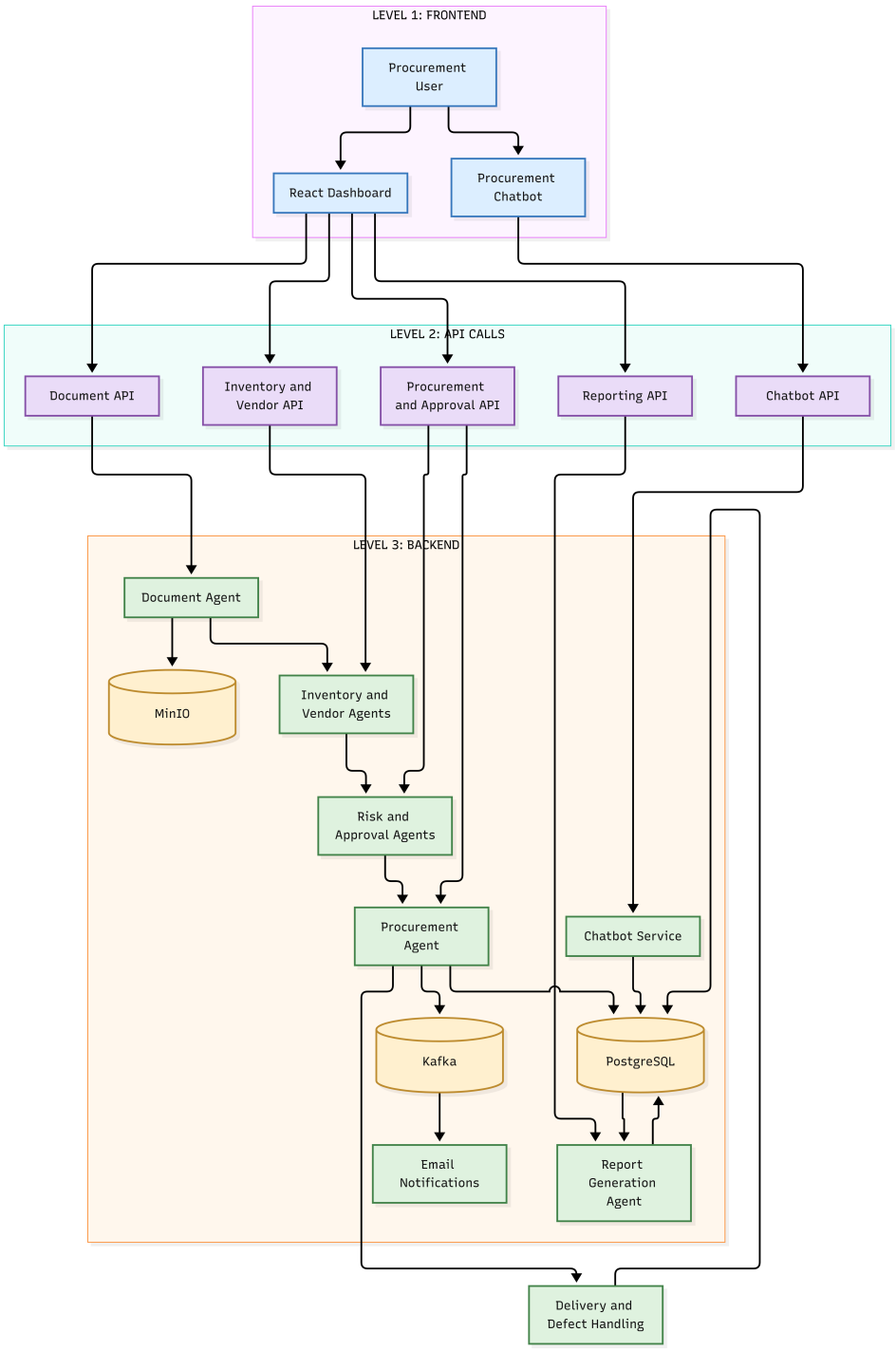


Figure 4.6: System Execution Flow Diagram

4.2.4.1 Level 1 – Frontend

The procurement user interacts with the React dashboard and Procurement Chatbot to submit requests, access procurement information, and obtain reports or chatbot responses.

4.2.4.2 Level 2 – API Calls

The API layer connects the frontend to the backend through dedicated Document, Inventory and Vendor, Procurement and Approval, Reporting, and Chatbot APIs. These APIs facilitate communication between the user interface and the corresponding backend services.

4.2.4.3 Level 3 – Backend

The backend processes requests through specialised agents, including the Document Agent, Inventory and Vendor Agents, Risk and Approval Agents, Procurement Agent, Chatbot Service, and Report Generation Agent. MinIO stores documents, PostgreSQL maintains structured procurement data, and Kafka supports event-driven communication and email notifications. The Delivery and Defect Handling component manages post-procurement activities related to delivery and defective items.

Overall, the architecture separates user interaction, API communication, and backend processing, enabling modular procurement automation, data management, chatbot interaction, and report generation.

4.2.5 Financial Approval Framework

After the purchase order is generated, the system determines the required approval level based on the total purchase amount. The approval framework contains three financial tiers.

Table 4.2: Financial Approval Tiers

Approval Tier	**Purchase Amount**	**Designated Approver**
TIER_1_OFFICER	$\leq$ Rs. 50,000	Procurement Officer
TIER_2_MANAGER	$>$ Rs. 50,000 and $\leq$ Rs. 2,00,000	Operations Manager
TIER_3_DIRECTOR	$>$ Rs. 2,00,000	Finance Director

The approval framework ensures that automated procurement execution does not remove financial governance. Routine purchases can be handled at the officer level, while higher-value purchases are escalated to managers or senior financial authorities.

4.2.6 Human-in-the-Loop Governance

The final stage of the framework incorporates human approval and audit controls. Once a PO is generated, the designated authority receives an approval notification. The approver can approve or reject the purchase order according to the assigned financial tier.

Approved purchase orders are digitally signed and stored, while the approval decision is recorded in the procurement audit ledger. This approach combines autonomous processing with human oversight and provides traceability for procurement decisions.

Chapter 5

Result and Discussion

5.1 Screenshots of the System

The graphical user interface of ProcureFlow provides an integrated platform for managing procurement requests, monitoring inventory, evaluating vendors, processing purchase orders, and tracking agent execution. The following screenshots demonstrate the major functionalities implemented in the system.

5.1.1 Connect API Data Source

The Data Sources interface enables administrators to connect external inventory and vendor databases using API keys.

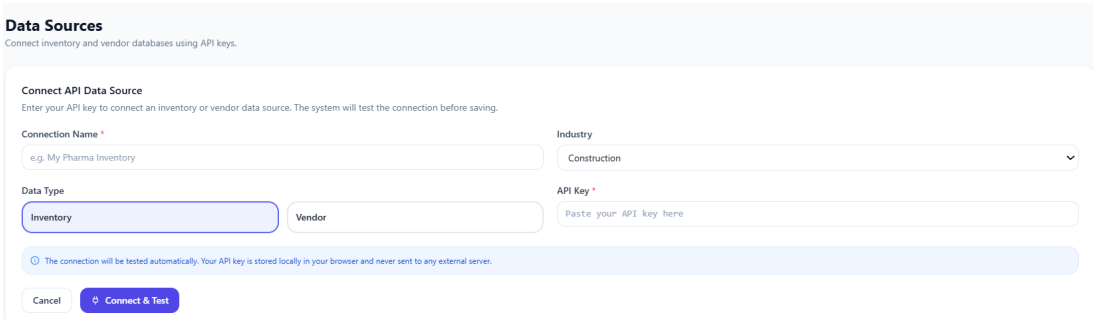


Figure 5.1: Connect API Data Source

5.1.2 Multi-Inventory Database

The inventory interface displays material details, available quantities, and stock information retrieved from connected inventory sources.

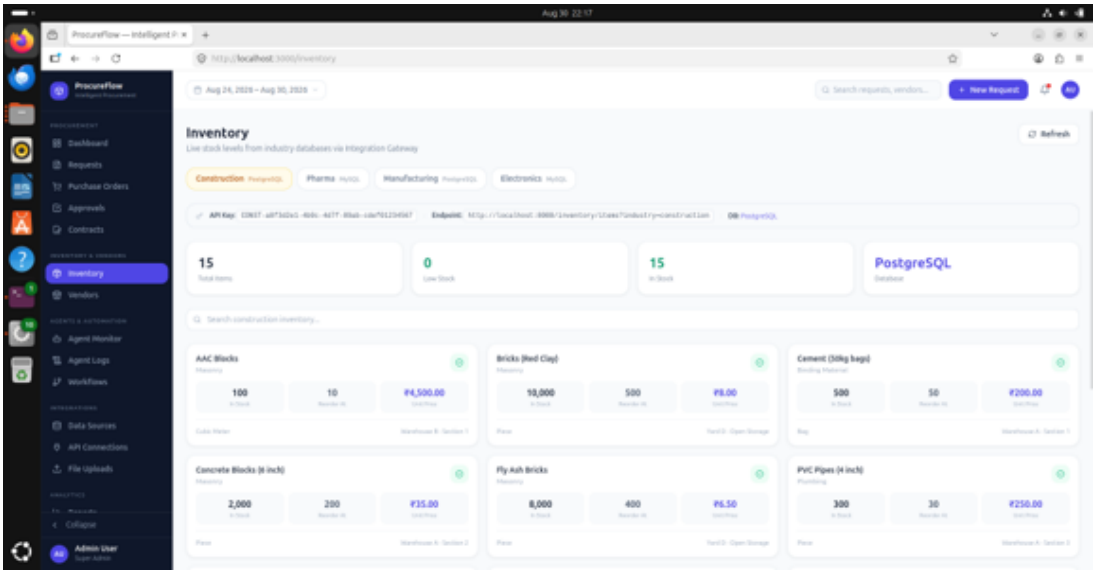


Figure 5.2: Multi-Inventory Database

5.1.3 Vendor Database

The Vendor Database interface displays supplier profiles, ratings, and product information used during vendor evaluation.

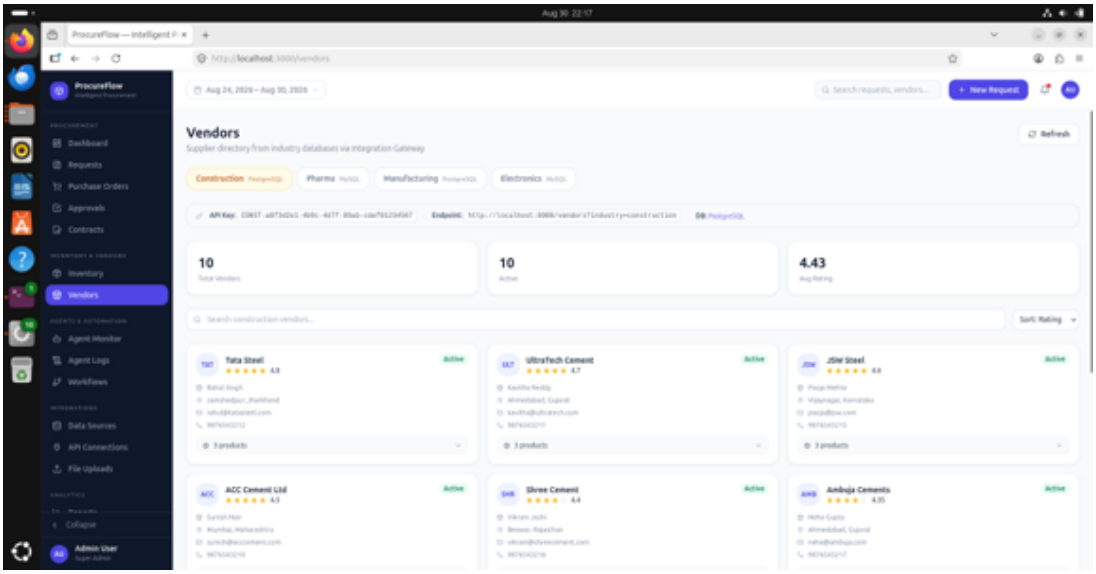


Figure 5.3: Vendor Database

5.1.4 Document Processing View

The document processing interface displays uploaded procurement documents and tracks their processing status.

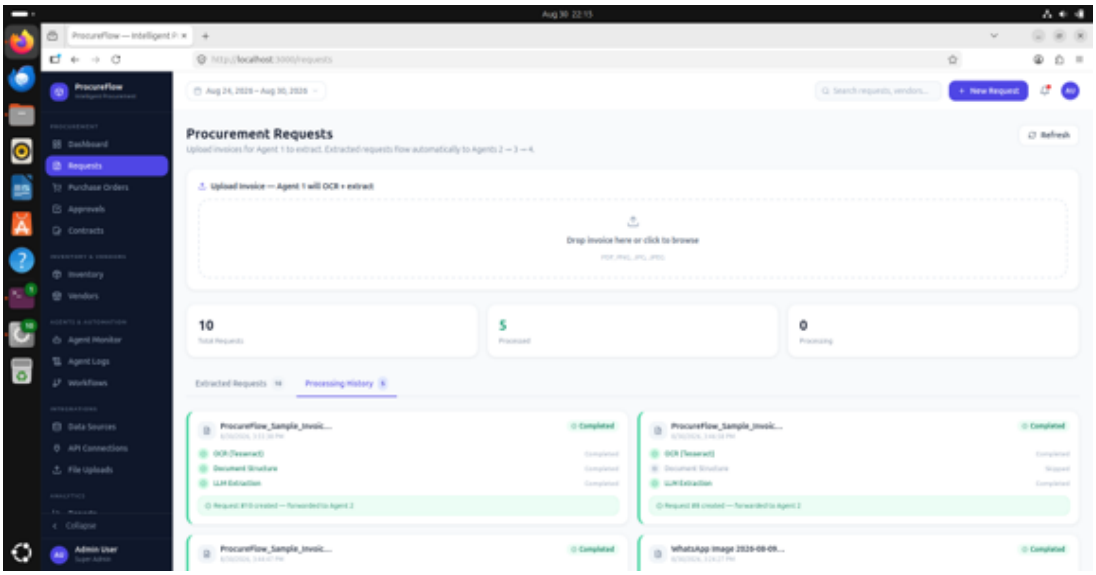


Figure 5.4: Document Processing View

5.1.5 Extracted Line Items

The system extracts item descriptions, quantities, and estimated costs from uploaded procurement documents using document intelligence techniques.

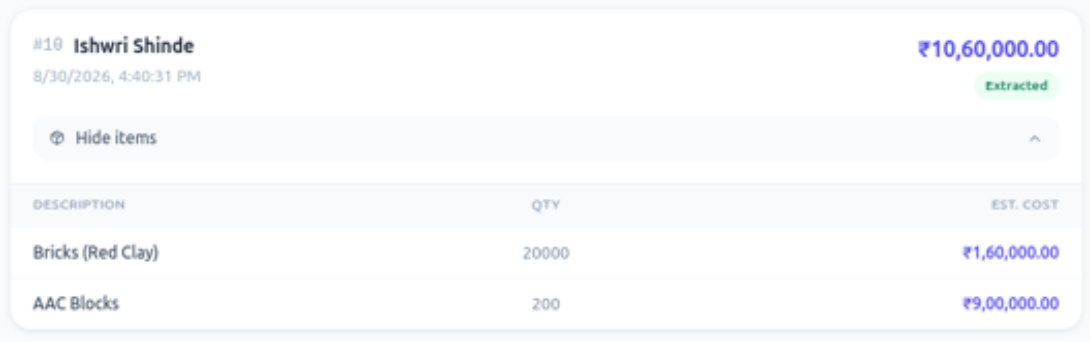


Figure 5.5: Extracted Line Items

5.1.6 Inventory Agent – Shortage Detection

The Inventory Agent compares required quantities with available stock and identifies material shortages that require procurement.

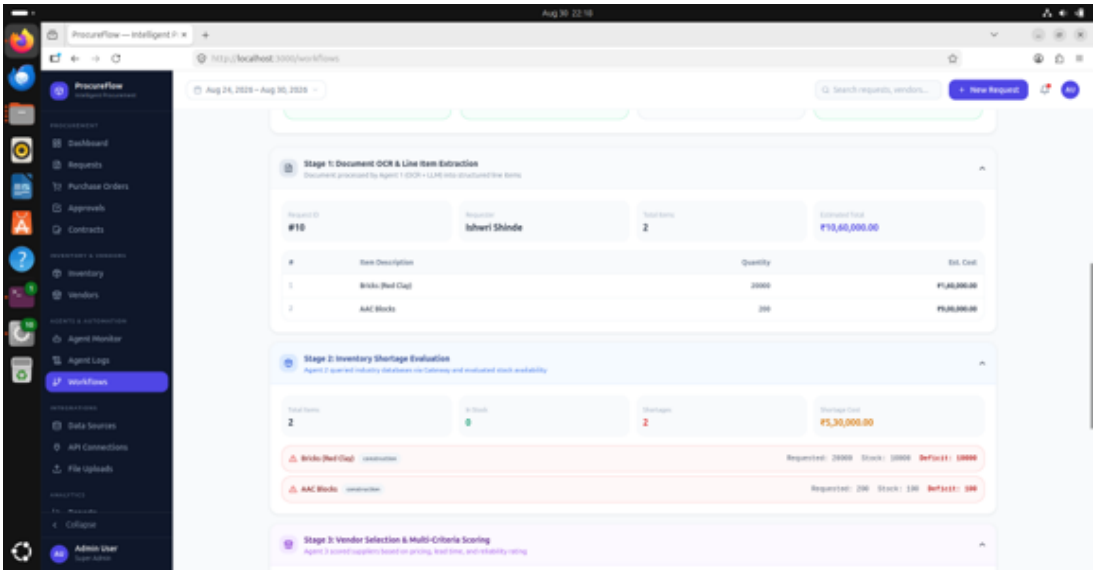


Figure 5.6: Inventory Agent – Shortage Detection

5.1.7 Vendor Recommendation

The Vendor Selection Agent evaluates eligible suppliers using pricing, delivery performance, quality indicators, and vendor-related risk information to recommend suitable vendors.

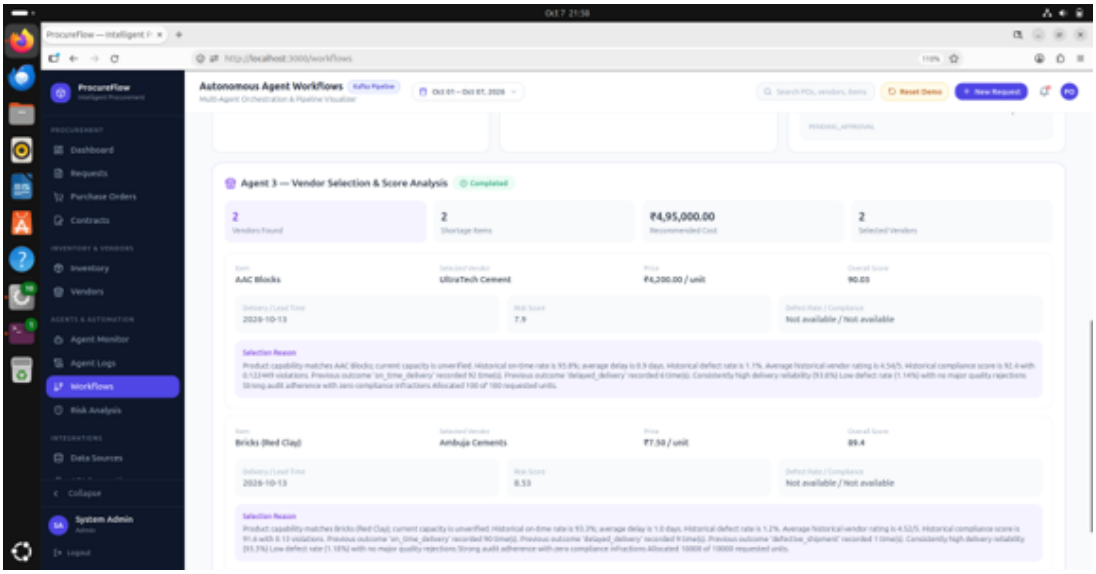


Figure 5.7: Vendor Recommendation and Score Analysis

5.1.8 Risk Analysis Agent

The Risk Analysis interface presents vendor risk scores and related evaluation results to support procurement decisions.

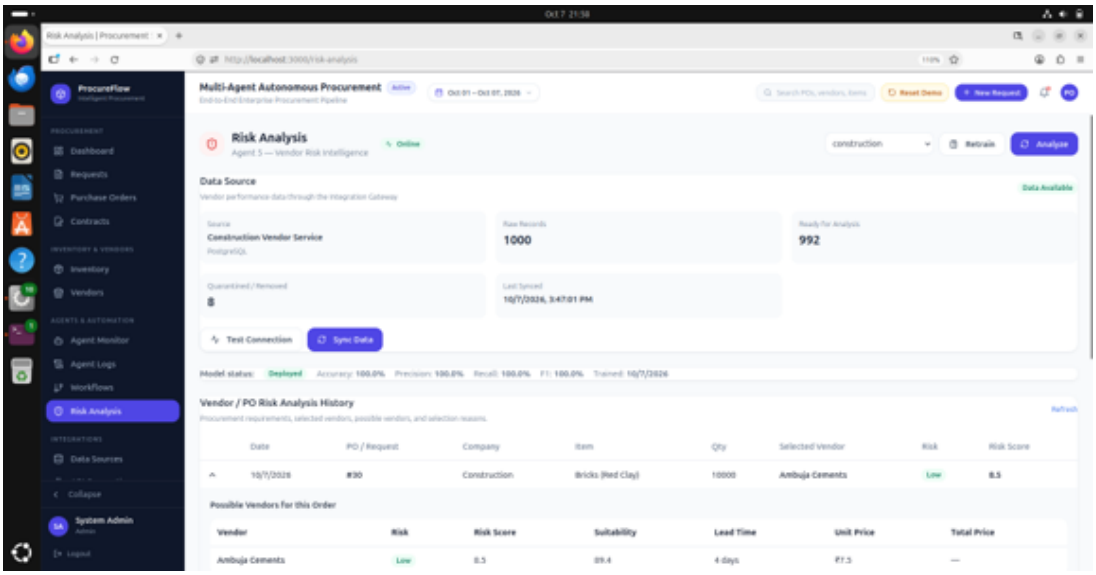


Figure 5.8: Risk Analysis Agent

5.1.9 Role-Based Approval Page

The approval interface allows authorised personnel to review purchase orders, make approval decisions, and manage digital signatures according to their assigned roles.

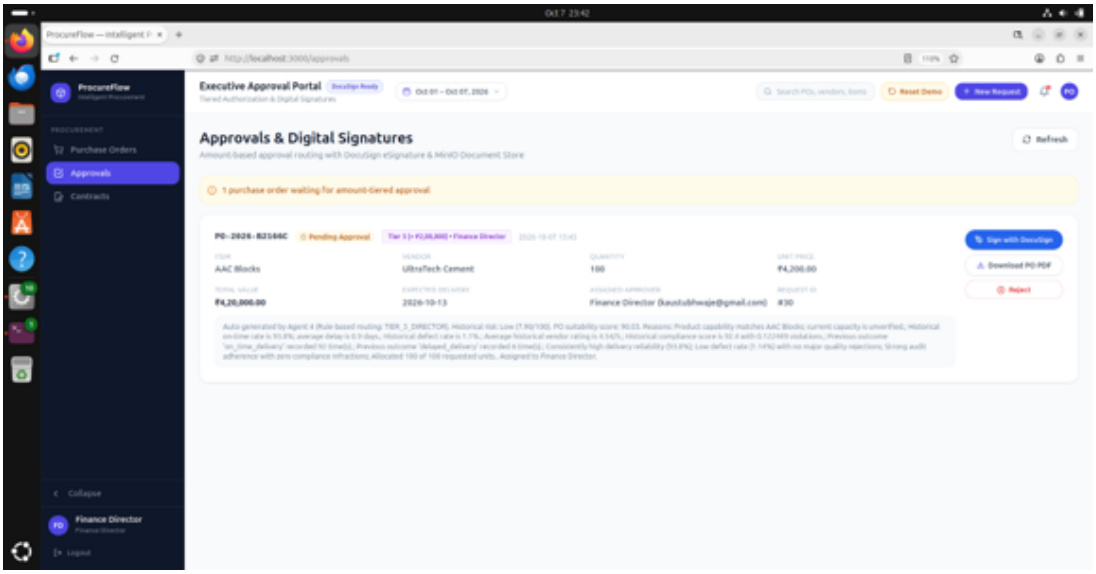


Figure 5.9: Role-Based Approval and Digital Signature

5.1.10 Purchase Order List

The Purchase Orders interface displays generated orders, supplier details, order values, delivery information, and approval statuses.

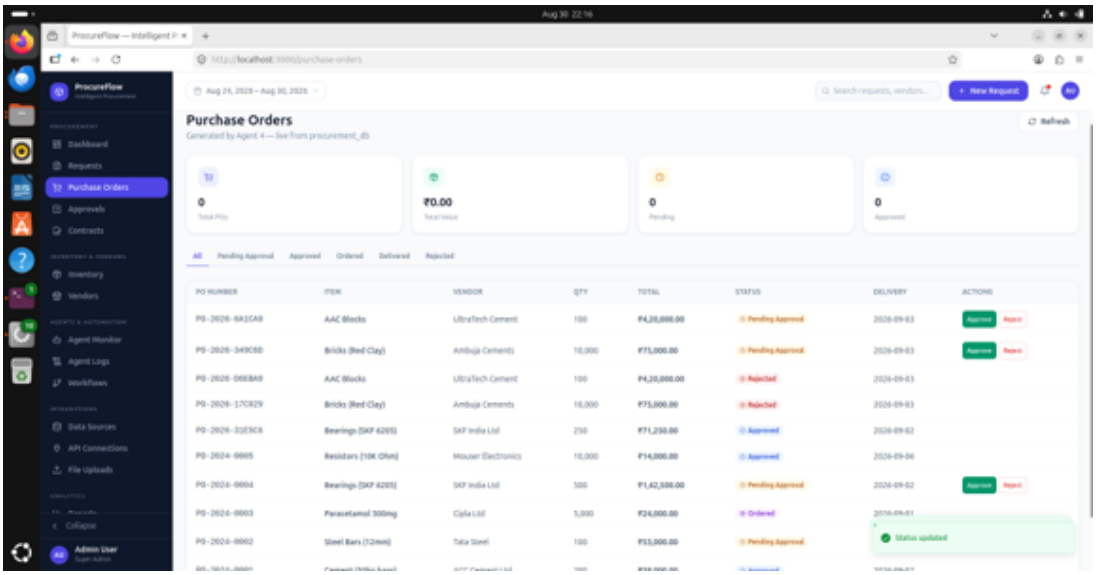


Figure 5.10: Purchase Order Management

5.1.11 Agent Workflow

The workflow dashboard provides a consolidated view of agent execution, inventory evaluation, vendor recommendations, and procurement processing status.

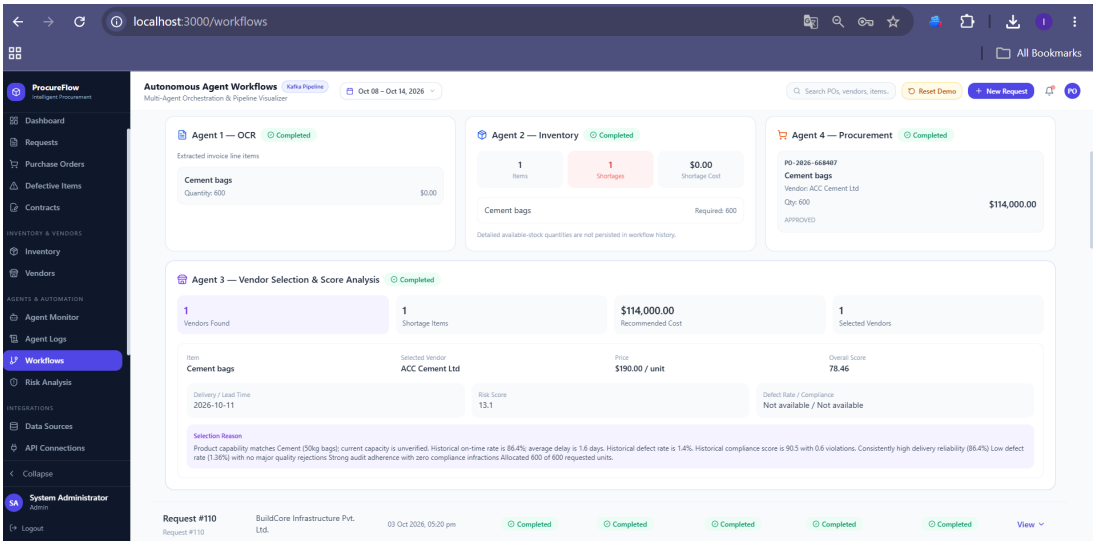


Figure 5.11: Autonomous Agent Workflow Dashboard

5.1.12 Kafka Logs

The Kafka Logs interface displays recorded procurement events, source agents, event types, Kafka topics, and event payloads for workflow monitoring and traceability.

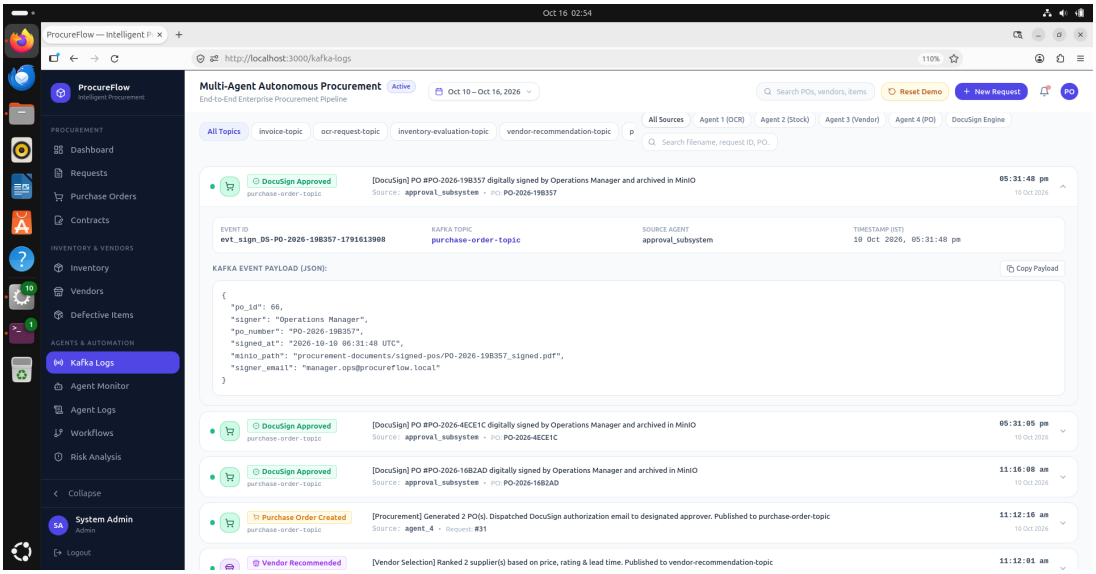


Figure 5.12: Kafka Logs and Event History

5.1.13 Purchase Order Approval Email Notification

The email notifies designated approvers about pending purchase orders and provides a link for digital-signature approval.

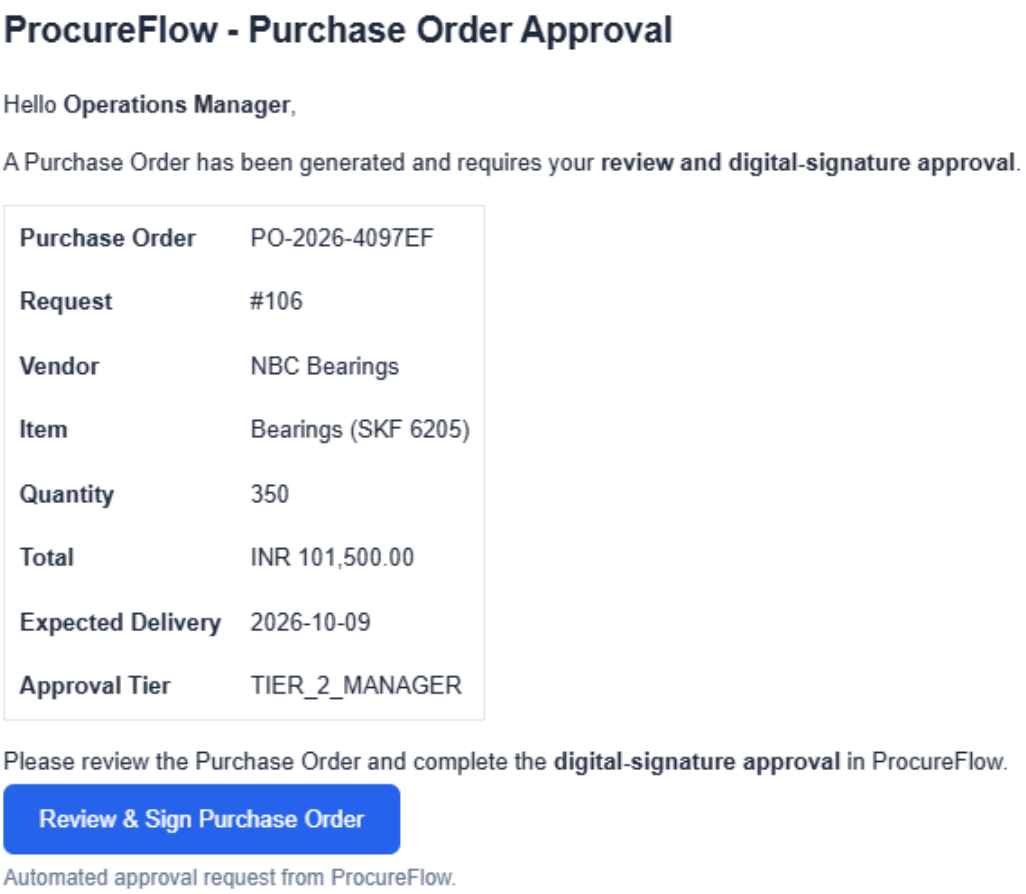


Figure 5.13: Role Based Purchase Order Email

5.1.14 Database API Calling

The API integration interface demonstrates the communication between the application and external data sources for retrieving inventory and vendor information.

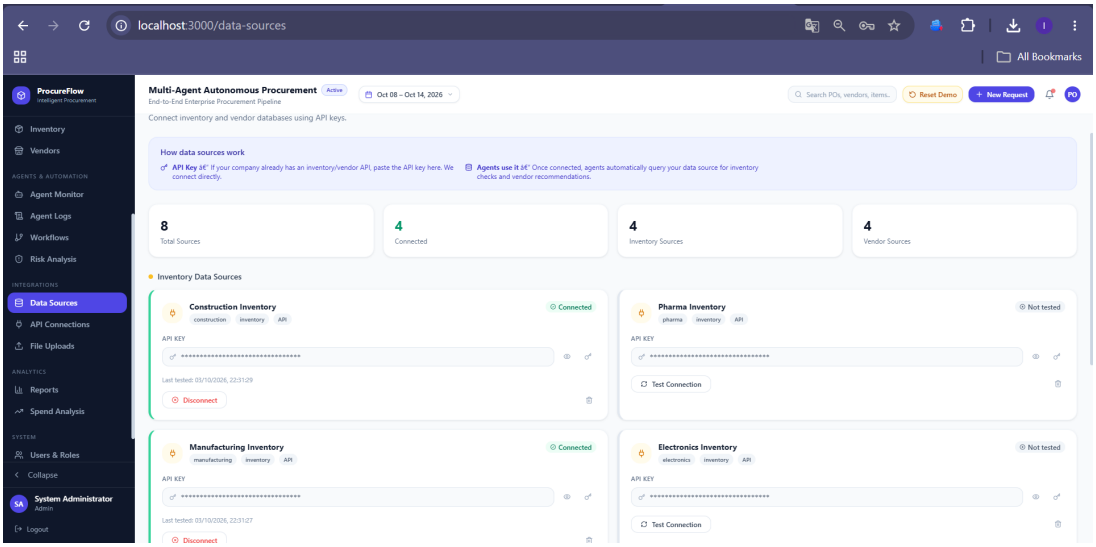


Figure 5.14: Database API Integration

5.2 Sample Code

5.2.1 Agent 1: OCR and Requisition Extraction Pipeline

```
@app.post("/extract-requisition")
async def extract_requisition_document(file: UploadFile = File(...)):
    """Extracts line items, vendor hints, and quantities from uploaded document."""
    file_bytes = await file.read()
    extracted_text = perform_ocr_extraction(file_bytes)

    # Structured Entity Extraction via LLM / Regex Parser
    parsed_requisition = extract_entities(extracted_text)

    # Publish to Kafka 'requisition-created' topic
    kafka_producer.send(
        topic="procurement.requisitions",
        key=str(parsed_requisition["request_id"]).encode("utf-8"),
        value=json.dumps(parsed_requisition).encode("utf-8")
    )
    return {"status": "QUEUED", "request_id": parsed_requisition["request_id"]}
```

Listing 5.1: OCR and Requisition Extraction

5.2.2 Agent 2: Inventory and Shortage Verification

```
def check_stock_and_reorder(item_name: str, requested_qty: int, db_cursor):
    """
    Evaluates current inventory levels against requested demand and reorder point.
    """
    db_cursor.execute("""
        SELECT id, name, quantity_in_stock, reorder_level
        FROM inventory_items
        WHERE LOWER(name) = LOWER(%s);
    """, (item_name,))
    item = db_cursor.fetchone()

    if not item:
        return {"status": "UNKNOWN_ITEM", "procure_needed": requested_qty}

    available_stock = item["quantity_in_stock"]
    shortage = max(0, requested_qty - available_stock)

    return {
        "item_id": item["id"],
        "in_stock": available_stock,
        "reorder_threshold": item["reorder_level"],
        "procure_needed": shortage,
        "is_shortage": available_stock <= item["reorder_level"]
    }
```

Listing 5.2: Inventory Shortage Verification

5.2.3 Agent 3: Vendor Selection and Multi-Factor Scoring

```
def rank_vendors(candidates: list, price_weight=0.4, lead_weight=0.3, quality_weight=0.3):
    """Computes weighted suitability score for candidate vendors."""
    ranked = []
    for vendor in candidates:
        norm_price = 1.0 / (1.0 + float(vendor["unit_price"]))
        norm_lead = 1.0 / (1.0 + float(vendor["lead_time_days"]))
        quality_score = float(vendor["compliance_score"]) / 100.0

        final_score = (
            (norm_price * price_weight) +
            (norm_lead * lead_weight) +
            (quality_score * quality_weight)
        )
        ranked.append({**vendor, "composite_score": round(final_score, 4)})

    return sorted(ranked, key=lambda x: x["composite_score"], reverse=True)
```

Listing 5.3: Vendor Ranking Algorithm

5.2.4 Agent 4: Purchase Order Generation and Approval Tiers

```
def determine_approval_tier(total_amount_inr: float) -> str:
    """Enforces multi-tier governance based on financial value."""
    if total_amount_inr < 50_000:
        return "TIER_1_OFFICER"       # Procurement Officer Sign-off
    elif total_amount_inr < 250_000:
        return "TIER_2_MANAGER"       # Operations Manager Sign-off
    else:
        return "TIER_3_DIRECTOR"      # Finance Director + Legal Sign-off


def create_purchase_order(po_payload: dict, db_cursor):
    tier = determine_approval_tier(po_payload["total_price"])
    db_cursor.execute("""
        INSERT INTO purchase_orders (
            po_number, request_id, vendor_id, vendor_name,
            item_name, quantity, unit_price, total_price,
            approval_tier, status
        ) VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, 'PENDING_APPROVAL')
        RETURNING id, po_number, status, approval_tier;
    """, (
        po_payload["po_number"], po_payload["request_id"], po_payload["vendor_id"],
        po_payload["vendor_name"], po_payload["item_name"], po_payload["quantity"],
        po_payload["unit_price"], po_payload["total_price"], tier
    ))
    return db_cursor.fetchone()
```

Listing 5.4: Purchase Order Generation and Approval

5.3 Testing

Software testing was performed to verify the functional correctness, reliability, security, and workflow integration of the proposed Agentic AI-based procurement system. The test cases cover the major stages of the system, including document processing, fallback mechanisms, inventory shortage detection, vendor filtering and ranking, purchase-order generation, financial approval, authorization, document storage, and industry isolation.

The testing was carried out using production-style procurement scenarios. Each test case was evaluated against an expected result and the actual system response was recorded. The test cases and their outcomes are presented below.

Table 5.1: Software Test Cases

ID	Test Case	Expected Result	Actual Result	Status
TC01	Invoice OCR and Extraction	All required fields and line items should be extracted from a 3-page invoice.	3 items extracted with 100% field extraction in 13.8 seconds.	PASS
TC02	LLM Fallback Processing	Gemini should be automatically used when the Groq service returns a rate-limit error.	Groq 429 error triggered fallback to Gemini successfully.	PASS
TC03	Inventory Shortage Detection	Shortage should be calculated as requested quantity minus available stock.	For 500 SKF bearings and stock of 120, shortage of 380 was detected.	PASS
TC04	Vendor MOQ Validation	A vendor whose MOQ exceeds the required quantity should be rejected.	Vendor with MOQ of 200 was rejected for a shortage quantity of 50.	PASS
TC05	Vendor Ranking	Vendor with the highest calculated composite score should receive the highest ranking.	Vendor A scored 0.89 and Vendor B scored 0.76; Vendor A ranked first.	PASS
TC06	Purchase Order Generation	A unique purchase-order number should be generated for the procurement workflow.	PO-2026-000108 was generated successfully.	PASS
TC07	Approval Tier Assignment	The system should assign the appropriate approval tier based on purchase value.	PO value of Rs. 1,42,500 was assigned to TIER_2_MANAGER.	PASS
TC08	Unauthorized Approval	A user without sufficient approval authority should not be allowed to approve the PO.	Officer attempting Tier-3 approval received HTTP 403.	PASS
TC09	PO PDF and Audit Storage	Generated PO should be converted to PDF, stored in MinIO, and its reference should be recorded in the audit ledger.	PDF was generated, signed, stored in MinIO, and the URL was persisted.	PASS
TC10	Industry Database Isolation	A procurement request for a disabled or unavailable industry database should be halted without accessing another industry's data.	Pharma request was halted with "Industry pharma disabled" and data leakage was prevented.	PASS

5.3.1 Test Case Descriptions

5.3.1.1 TC01 – Invoice OCR and Structured Extraction

The first test case verifies the Document Intelligence Agent. A 3-page procurement invoice containing multiple line items was provided as input. The document was uploaded and processed through MinIO and Tesseract OCR, followed by LLM-based structured extraction.

The system successfully extracted the required fields and three line items. The test achieved 100% field extraction within 13.8 seconds. Therefore, the test confirms that the document-processing pipeline can convert an unstructured procurement document into structured procurement information.

5.3.1.2 TC02 – LLM Fallback Mechanism

This test case verifies the fallback mechanism used during AI-based document processing. A rate-limit condition was simulated for the primary Groq processing service.

When the Groq service returned an HTTP 429 response, the system automatically switched to the Gemini processing service. The document-processing workflow continued successfully without requiring manual intervention. The test therefore confirms the resilience of the LLM processing component.

5.3.1.3 TC03 – Inventory Shortage Detection

This test verifies the Inventory Intelligence Agent. A procurement requirement for 500 SKF bearings was evaluated against an available inventory of 120 units.

The shortage was calculated as:

$$\text{Shortage} = \text{Requested Quantity} - \text{Available Stock}$$

$$\text{Shortage} = 500 - 120 = 380$$

The system correctly identified a shortage of 380 units and forwarded the shortage information for vendor evaluation. The test confirms the correctness of the inventory reconciliation process.

5.3.1.4 TC04 – Vendor MOQ Validation

This test verifies the vendor eligibility filtering mechanism. A shortage quantity of 50 units was provided while a candidate vendor had a minimum order quantity of 200 units.

Since the vendor MOQ was greater than the required quantity, the vendor was excluded from the eligible vendor list. This confirms that the system applies the predefined MOQ constraint before performing vendor ranking.

5.3.1.5 TC05 – Multi-Criteria Vendor Ranking

This test verifies the vendor scoring algorithm. Two vendors were evaluated using their rating, price, and delivery lead time.

Vendor A received a composite score of 0.89, while Vendor B received a score of 0.76. Since Vendor A achieved the higher final score, it was ranked first by the system.

The test confirms that the multi-criteria scoring framework correctly combines vendor rating, price, and delivery performance to generate the final ranking.

5.3.1.6 TC06 – Purchase Order Generation

This test verifies the Procurement Execution Agent. After successful inventory and vendor processing, the system generated a unique purchase order for the procurement request.

The generated purchase-order identifier was:

```
PO-2026-000108
```

The test confirms that the purchase-order generation process produces a unique identifier and successfully initiates the subsequent approval workflow.

5.3.1.7 TC07 – Financial Approval Tier Assignment

This test verifies the financial approval framework. A purchase order with a total value of Rs. 1,42,500 was submitted for approval.

According to the configured approval rules, purchases greater than Rs. 50,000 and less than or equal to Rs. 2,00,000 require manager-level approval. The system therefore assigned:

```
TIER_2_MANAGER
```

The request was routed to the appropriate manager-level approval process. This confirms that approval tiers are assigned according to purchase value.

5.3.1.8 TC08 – Unauthorized Approval Attempt

This test verifies role-based authorization. An officer-level user attempted to approve a purchase order requiring a higher approval tier.

The system rejected the request and returned an HTTP 403 authorization response. The purchase order was therefore not approved by the unauthorized user.

This test confirms that financial approval cannot be bypassed by users who do not possess the required authorization level.

5.3.1.9 TC09 – Purchase Order PDF and Audit Persistence

This test verifies the document-generation and audit-storage functionality. After successful purchase-order generation, the system created the PO as a PDF document using the configured document-generation process.

The generated PDF was digitally signed and stored in MinIO. The corresponding document reference URL was also persisted in the procurement audit ledger.

The successful execution of this test confirms that purchase-order documents and their associated audit information are retained for future reference and traceability.

5.3.1.10 TC10 – Industry Database Isolation

The final test verifies industry-level isolation and controlled database access. A procurement request associated with the pharmaceutical industry was submitted while the corresponding Pharma database was disconnected or disabled.

The system halted the request and recorded the message:

```
Industry pharma disabled
```

The request did not access data belonging to another industry. This confirms that the system applies industry-aware routing and prevents unintended cross-industry data access.

5.3.2 Test Summary

All ten functional test cases were successfully completed. The testing covered the complete procurement workflow, beginning with document ingestion and continuing through inventory evaluation, vendor recommendation, purchase order generation, financial approval, and audit persistence.

The results demonstrate that the implemented system performs the intended operations correctly under the evaluated scenarios. The tests also verify important reliability and governance mechanisms, including LLM fallback, vendor quality gates, approval authorization, digital document storage, and industry-level data isolation.

Table 5.2: Overall Software Testing Summary

Testing Parameter	**Result**	**Status**
Total Test Cases	10	Completed
Passed Test Cases	10	PASS
Failed Test Cases	0	–
Overall Test Result	100% Test Cases Passed	PASS

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

In conclusion, the project, Agentic AI for Smart Procurement in Manufacturing Industries, demonstrates an intelligent procurement workflow using specialized agents for document processing, inventory analysis, vendor evaluation, and procurement execution. By integrating OCR, LLM-based extraction, Apache Kafka, PostgreSQL, Redis, and MinIO, the system automates key procurement activities and reduces manual effort.

The Vendor Chatbot supports vendor queries and price negotiation, while the Report Generation Agent facilitates the generation of procurement-related reports and insights. The system also incorporates vendor scoring, purchase order generation, tier-based financial approvals, digital signatures, and audit tracking to ensure accountability and controlled decision-making.

Testing validated key system functionalities, demonstrating the potential of AI-driven automation to improve procurement efficiency, traceability, and decision support while maintaining human oversight. Overall, the project illustrates how agentic AI can streamline procurement operations in manufacturing industries while retaining approval controls and accountability.

6.2 Future Scope

The following enhancements can be considered for future development of the proposed system:

1. **Payment Gateway Integration:** Integrate a secure payment gateway to facilitate vendor payments after purchase order approval and fulfillment verification. This enhancement can include transaction tracking and automated payment status updates.

2. **Enhanced Vendor Chatbot:** Extend the chatbot with multilingual support, context-aware conversations, and improved negotiation capabilities to enable more effective communication with vendors.

3. **Automated Quality Inspection and Defect Tracking:** Integrate AI-based quality inspection to identify defective materials, maintain supplier defect records, and incorporate product quality performance into vendor evaluation.

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Acknowledgement

The success of a project like this, involving high technical expertise, patience, and massive support from guides, is possible when team members work together. We take this opportunity to express our gratitude to those who have been instrumental in the successful completion of this project. We would like to show our appreciation to **Prof. Nupur Gaikwad** for their tremendous support and help; without them, this project would not have reached fruition. We would also like to thank our project coordinator **Dr. Rakhi Kalantri** for providing us with regular inputs about documentation and project timeline. A big thanks to our HOD **Dr. M.Kiruthika** for all the encouragement given to our team. We would also like to thank our principal, **Dr. S. M. Khot**, and our college, **Fr. C. Rodrigues Institute of Technology, Vashi**, for giving us the opportunity and the environment to learn and grow.

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Appendix A : Timeline Chart

Project Schedule (GANTT CHART)

The following Gantt chart illustrates the timeline for the development of the System. It outlines the key phases of the project, their sequence, and respective durations to ensure smooth and structured execution.

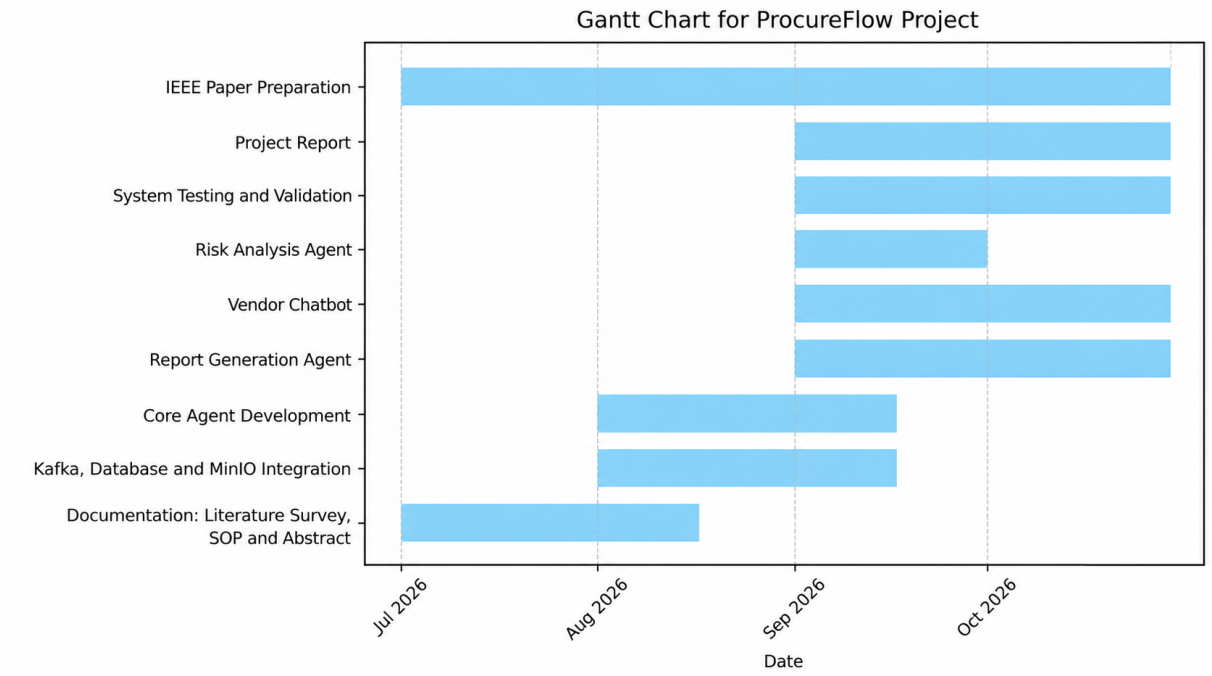


Figure 6.1: Gantt Chart for ProcureFlow Project

Project Timeline Explanation

The table summarizes the major project phases, their durations, timelines, and key activities undertaken during the development

Table 6.1: Project Timeline

Phase	Duration	Timeline	Description
Documentation and Requirement Analysis	~2 months	July–August 2026	Prepared the literature survey, SOP, abstract, and requirements.
Infrastructure Setup and Integration	~2 months	August–September 2026	Configured Kafka, PostgreSQL, Redis, and MinIO.
Core Agent Development	~2 months	August–September 2026	Developed the four core procurement agents.
Advanced Agent Development	~2 months	September–October 2026	Developed the Risk Analysis Agent, Vendor Chatbot, and Report Generation Agent.
Approval Workflow Integration	~2.5 months	Mid-August–October 2026	Integrated role-based approvals, email notifications, and digital signatures.
Testing and Validation	~2 months	September–October 2026	Tested system functionality, integration, and approval workflows.
Project Report Preparation	~2 months	September–October 2026	Documented implementation, screenshots, and testing results.
IEEE Paper Preparation	~4 months	July–October 2026	Prepared the research paper covering methodology and evaluation results.

Appendix B

The following Sustainable Development Goals (SDGs) are relevant to ProcureFlow. The mapping links the project's procurement automation features with industrial innovation, responsible resource use, and productivity.

SDG Mapping of the Project

Sr. No.	SDG No.	Complete Name of SDG	Rationale / Contribution of the Project
1	9	Industry, Innovation and Infrastructure	ProcureFlow combines AI agents, document processing, inventory checks, vendor evaluation, and purchase-order workflows to support digital innovation in manufacturing procurement.
2	12	Responsible Consumption and Production	Inventory checks can identify existing stock before new purchases. Vendor evaluation and defective-item tracking support better sourcing decisions and material traceability. Actual waste reduction would require deployment evidence.
3	8	Decent Work and Economic Growth	Automation of document extraction, stock verification, vendor comparison, purchase-order preparation, and reporting can reduce repetitive administrative work. Financial approval tiers retain human oversight.

_____

Signature of Project Guide

Name of Project Guide: _____

Date: _____

Appendix C

Appendix C: Complex Engineering Problem Identification and Evaluation

COMPLEX ENGINEERING PROBLEM IDENTIFICATION AND EVALUATION

The Washington Accord emphasizes the ability of engineering graduates to identify, analyse, and solve complex engineering problems using appropriate engineering knowledge, modern tools, multidisciplinary thinking, and analytical reasoning. The Major Project provides an opportunity for students to demonstrate these abilities through a structured and evidence-based project.

This appendix provides a structured mechanism to assess the complexity of the proposed Major Project. The assessment is initially carried out by the student team and subsequently verified by the Project Guide.

A. Project Details

Project Title	Agentic AI for Smart Procurement in Manufacturing Industries
Group / Team No.	42
Name of Student(s)	Kaustubh Waje; Ashley Fernandes; Dikshant Phulpagar; Ishwari Shinde
Project Guide	Mrs. Nupur Gaikwad
Academic Year	2026–2027

B. Complexity Evaluation

For each parameter, the student team shall assign a score from **0 to 3** and provide a suitable justification or evidence. The Project Guide shall review the student's assessment and record the verified score in the final column.

No.	Complexity Parameter	Student Score	Student Justification / Evidence	Guide Score
1	Problem novelty and research depth	2	The project combines document extraction, inventory checks, vendor evaluation, risk analysis, purchase-order generation, and approval support in one procurement workflow. Evidence may include the literature survey, problem definition, and completed modules.	
2	Multidisciplinary knowledge requirement	3	The project brings together AI/OCR, chatbot or RAG functionality, database design, web development, API integration, and procurement processes. Include only components that are part of the actual implementation.	
3	System design complexity	3	The proposed design separates frontend interaction, APIs, specialised agents, and data stores. The workflow coordinates document processing, inventory and vendor analysis, purchase-order generation, approvals, notifications, and reporting.	
4	Technology integration	2	The design identifies application services alongside PostgreSQL, Redis, Kafka, MinIO, and AI-related components. Indicate which technologies were configured and tested, rather than listing proposed technologies as completed work.	
5	Data analysis and experimentation	2	Inventory quantities, vendor prices and availability, quality information, and delivery-performance data can inform procurement decisions. Supporting evidence may include test cases, sample records, comparisons, and recorded results.	
6	Societal / industrial relevance	3	The project addresses manufacturing procurement activities such as document handling, stock verification, supplier comparison, approval routing, and procurement traceability. Evidence may include requirements analysis and a system demonstration.	
7	Stakeholder involvement	2	The workflow considers procurement users, vendors, and designated approvers through vendor interaction, notifications, and approval tiers. Clarify whether these roles were validated through stakeholder consultation or considered during system design only.	
8	Sustainability considerations	2	Inventory-aware purchasing and defective-item tracking may support better material decisions and traceability. Do not claim measured reductions in waste, cost, or resource use unless supporting evidence is available.	
	Total Score	**19/24**		**/24**

Scoring Guideline

Score	Interpretation
0	Not applicable / negligible
1	Basic level of complexity
2	Moderate level of complexity
3	Significant / high level of complexity

Maximum Score: 24

C. Final Project Classification

The final classification shall be based on the score verified by the Project Guide.

Score Range	Project Classification
0–8	Simple Engineering Problem
9–16	Moderately Complex Engineering Problem
17–24	Complex Engineering Problem

Final Classification:

Simple Engineering Problem

Moderately Complex Engineering Problem

Complex Engineering Problem

Project Guide Signature

Date: _____

Appendix D: Project Outcome

Add publication details, project competition participation/winning